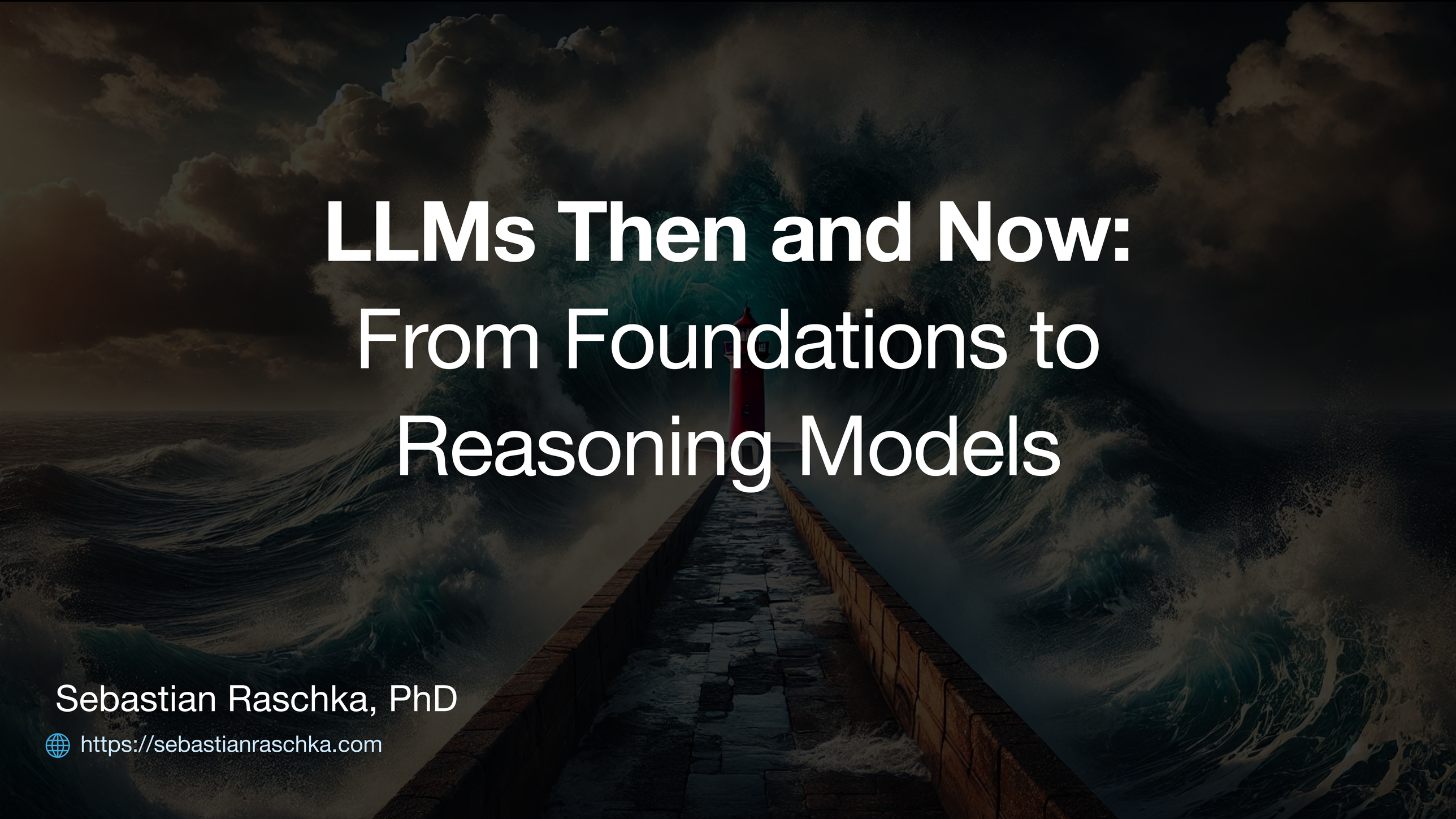


LLMs Then and Now: From Foundations to Reasoning Models

Sebastian Raschka, PhD

https://sebastianraschka.com

Topics of this talk

1

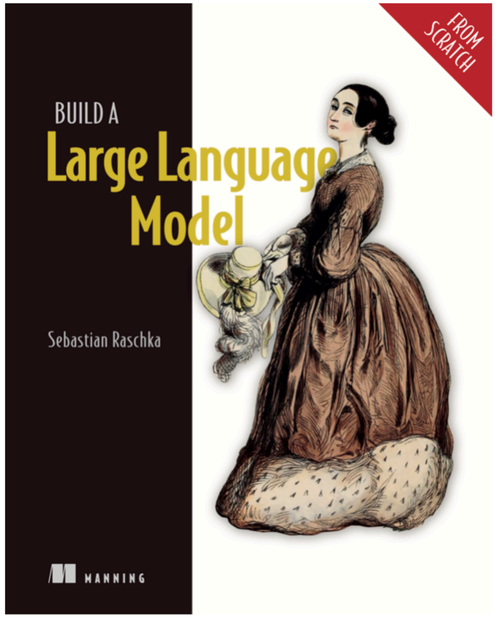
Fundamentals

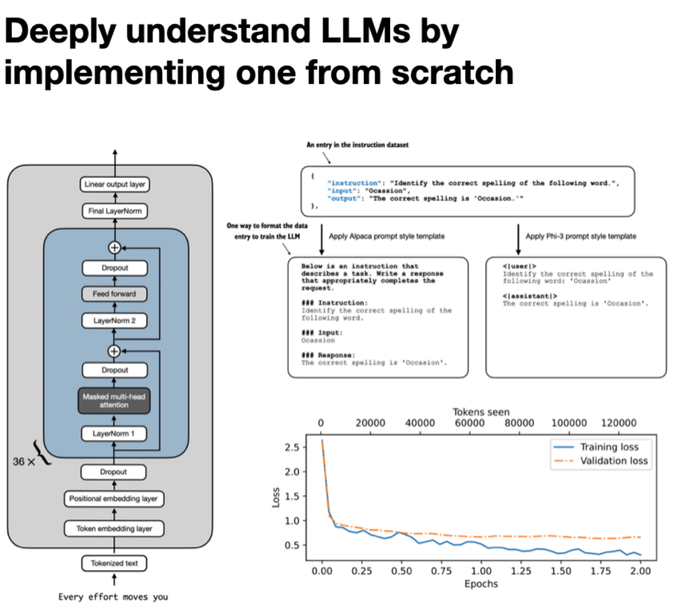
How LLMs work

2

What's new & what's coming?

E.g., reasoning models





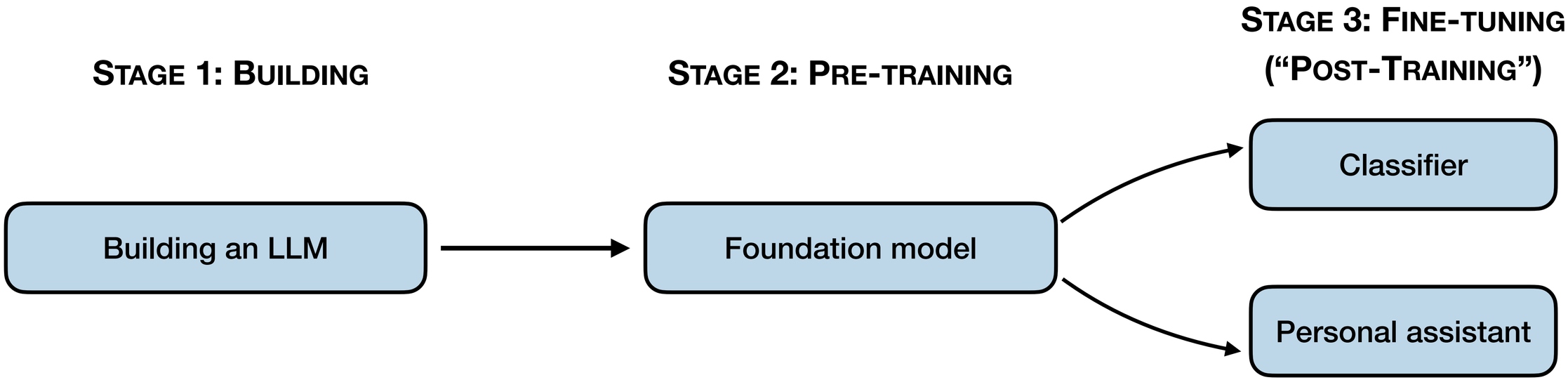
Sebastian Raschka, PhD

https://sebastianraschka.com

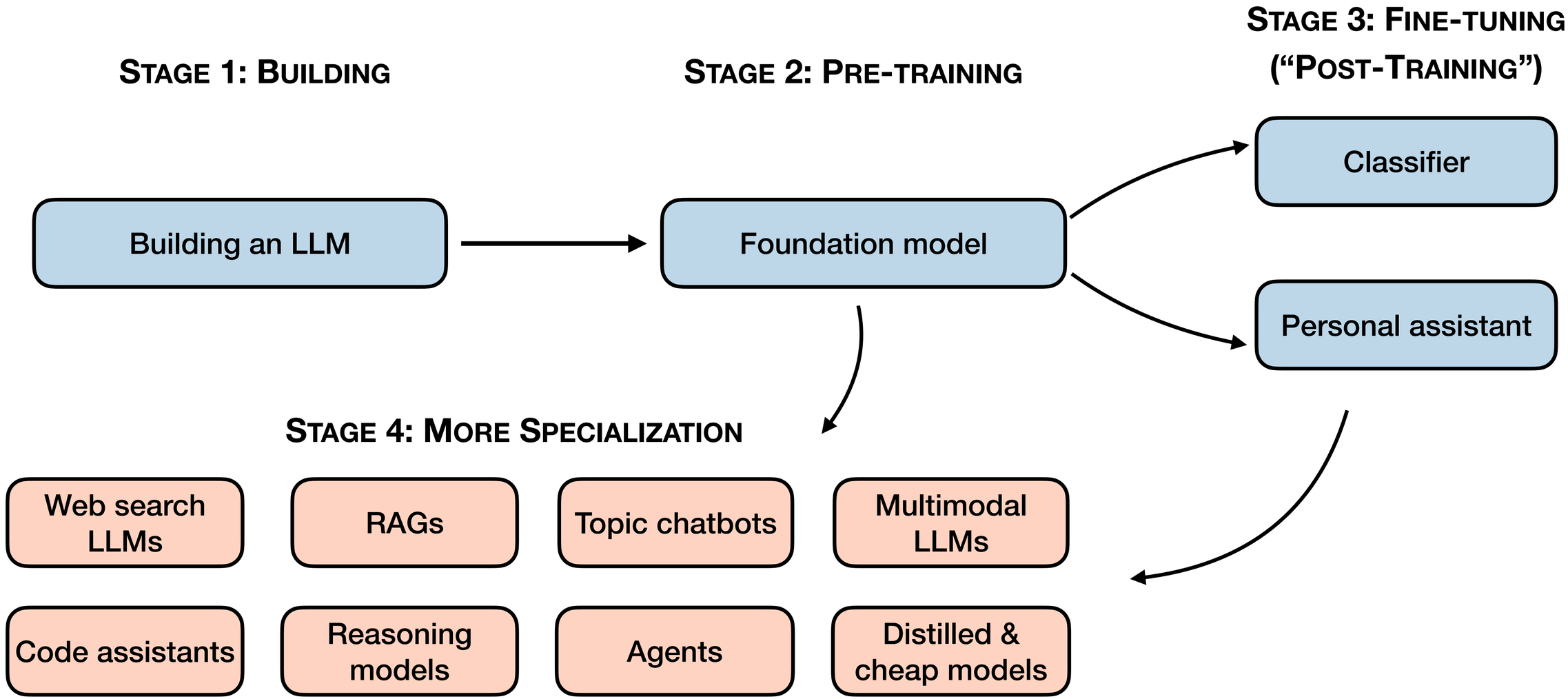
https://github.com/rasbt/LLMs-from-scratch

What goes into developing Large Language Models (LLMs)?

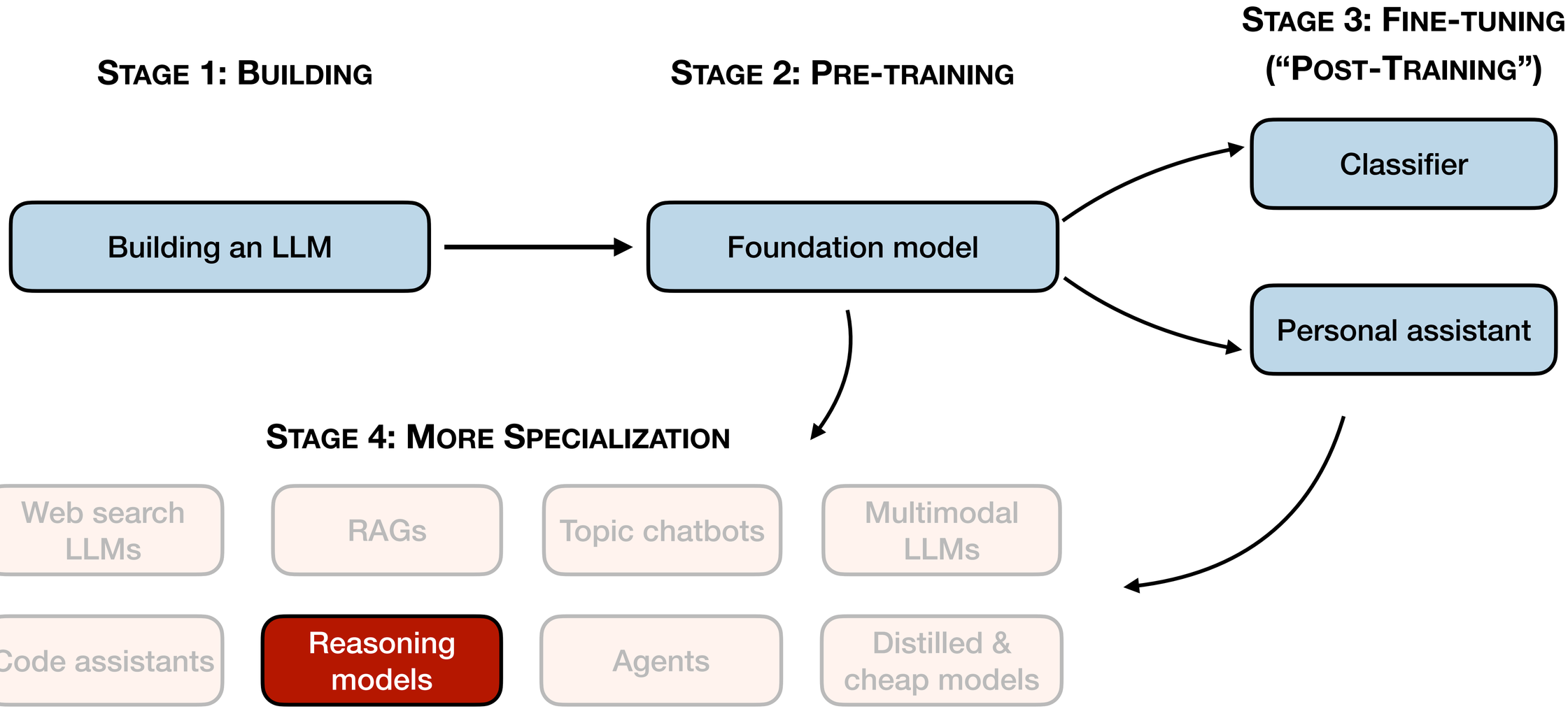
Developing an LLM



Developing an LLM

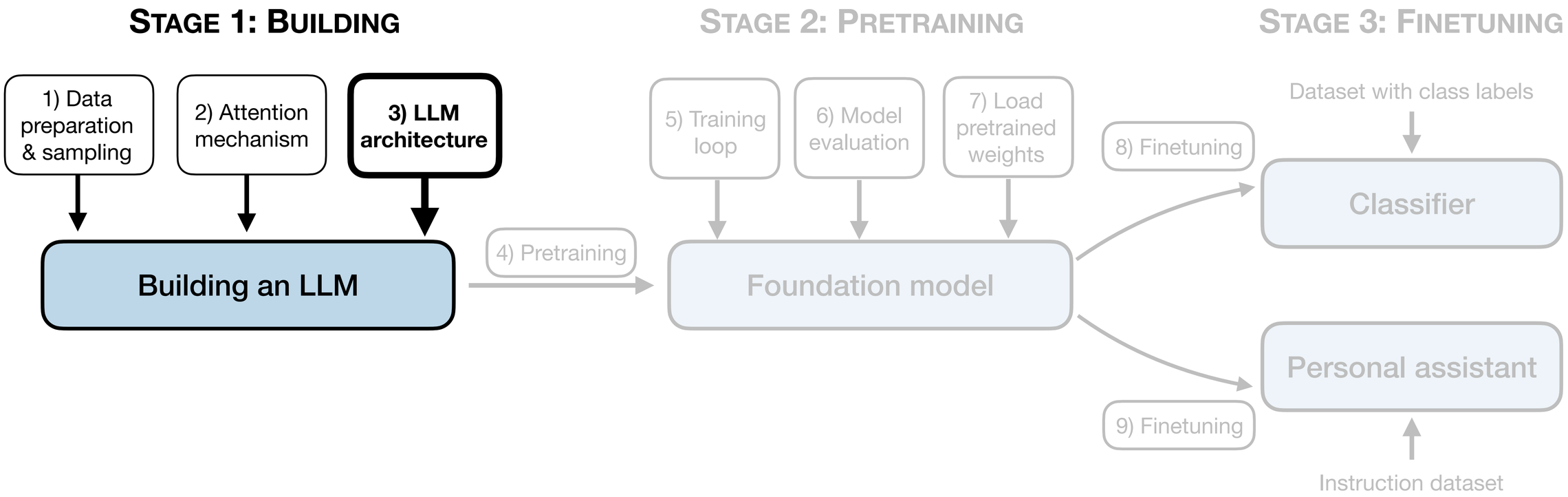


Developing an LLM

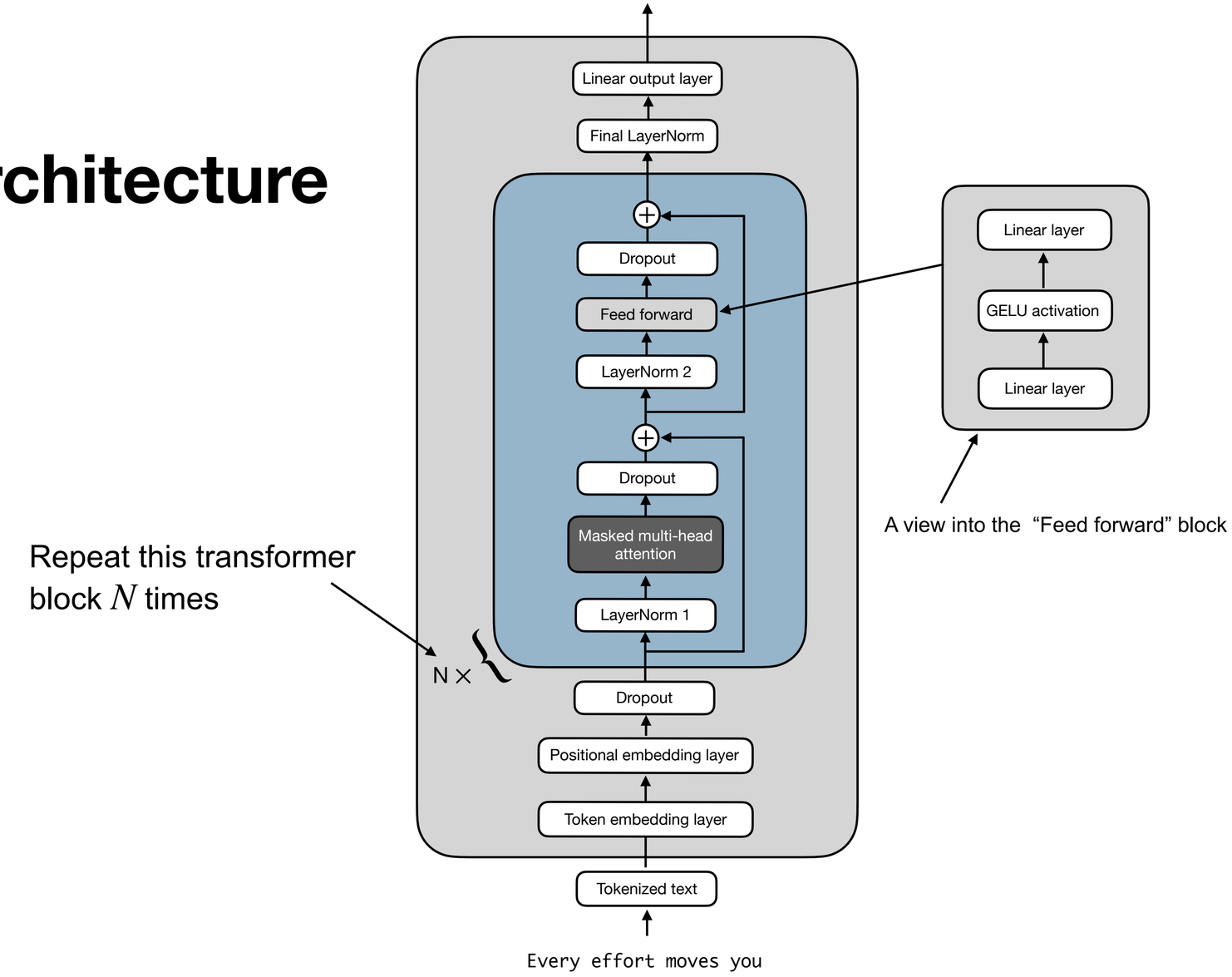


Stage 1: Building

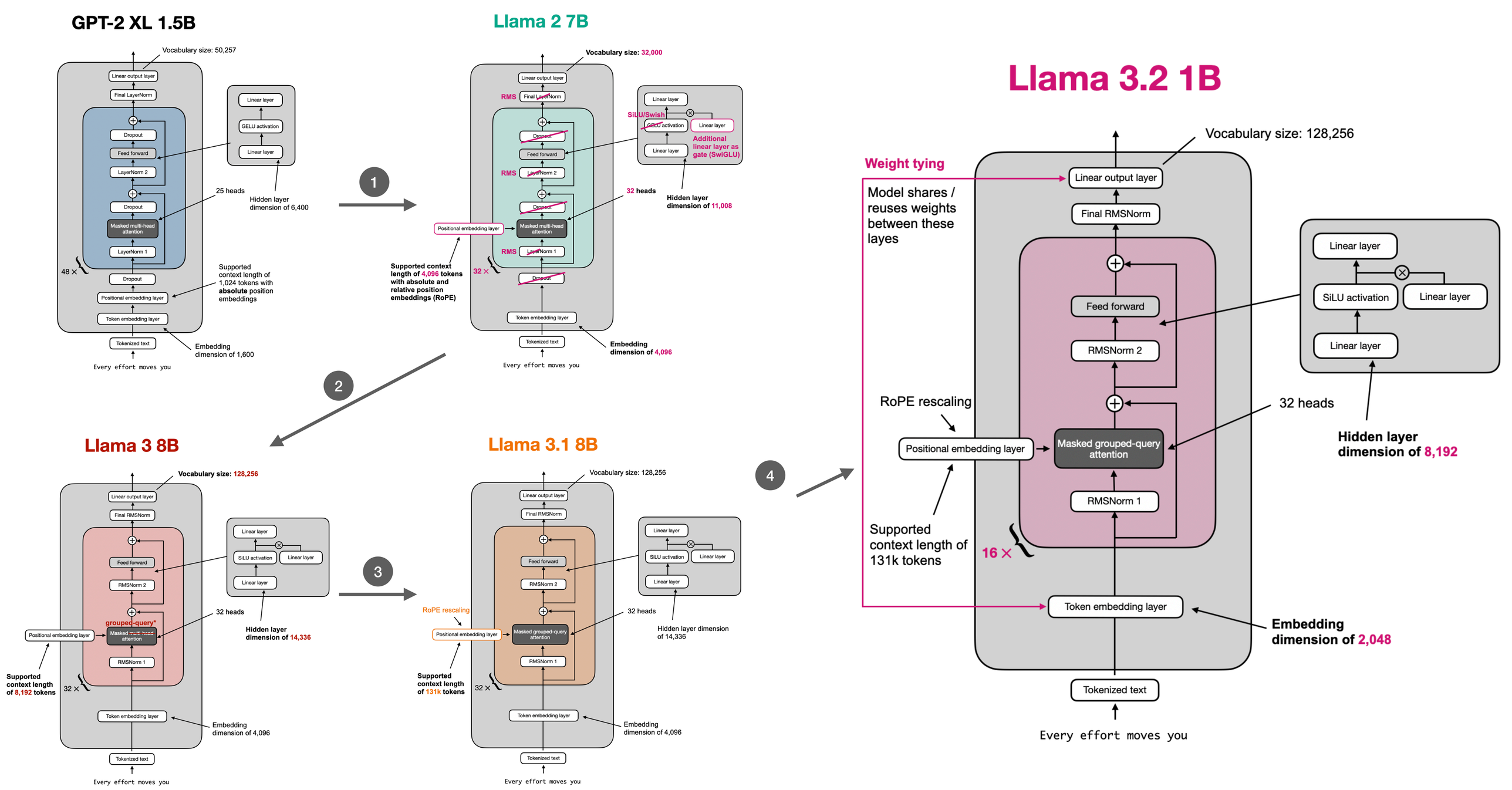
Implementing the architecture



Original GPT architecture (2018)



Not “that much” has changed



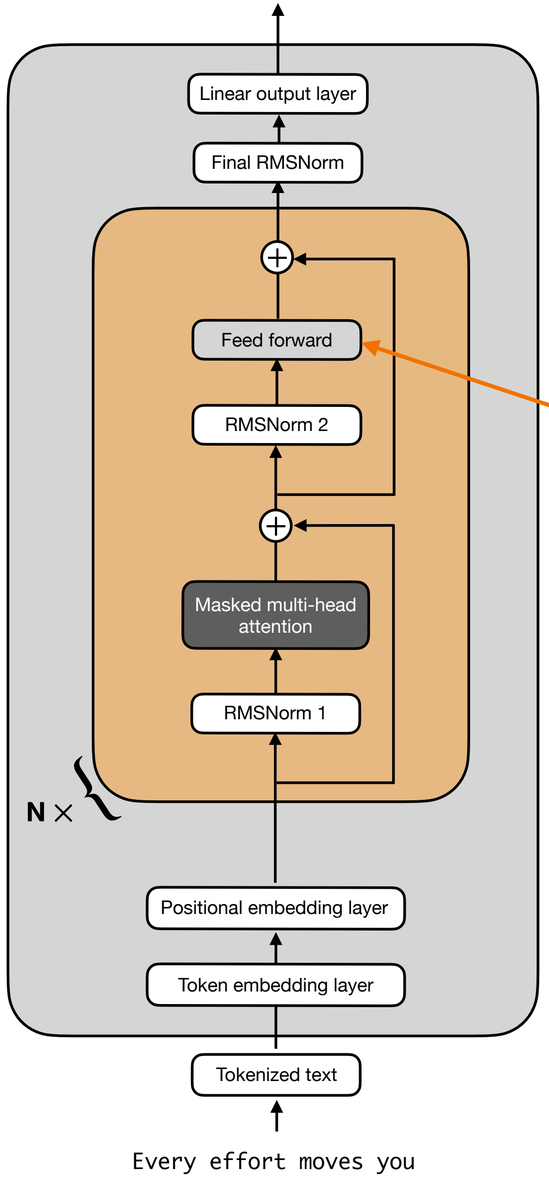
* The larger Llama 2 34B and 70B also used grouped-query attention

Sebastian Raschka

https://github.com/rasbt/LLMs-from-scratch/tree/main/ch05/07_gpt_to_llama

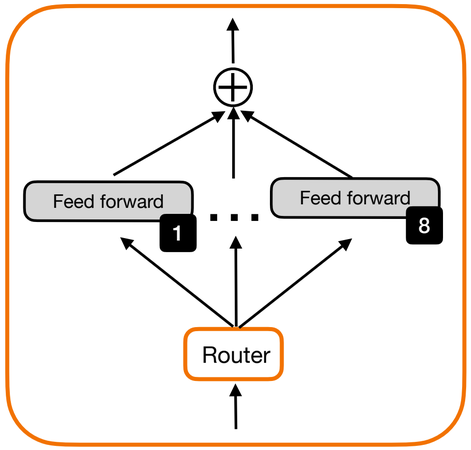
Takeaway: Not “that much” has changed

Mixture of Experts



Mixtral (Jan 2024)

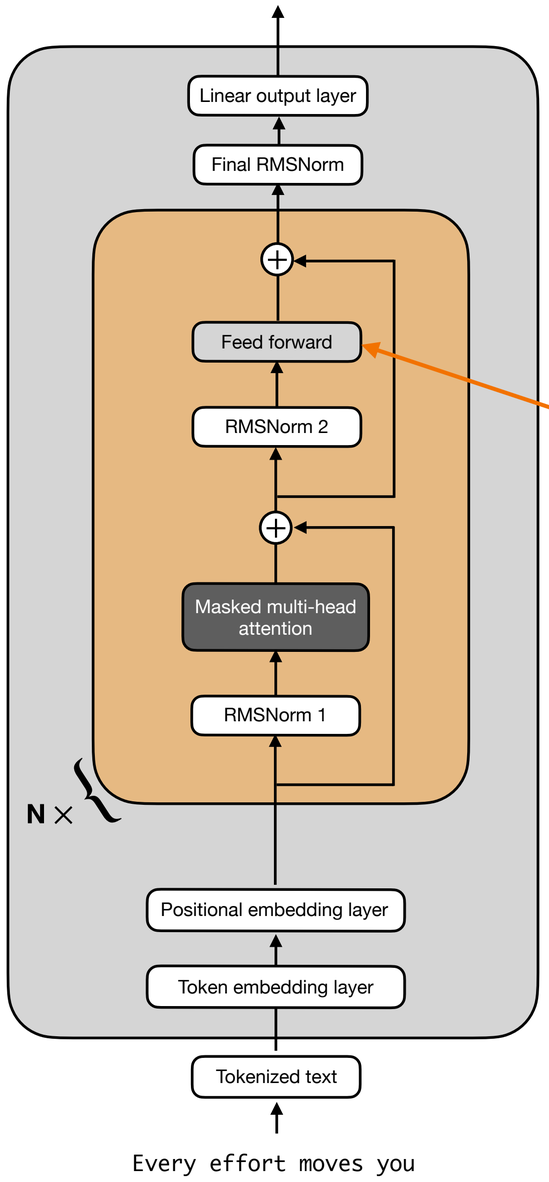
replaces the feedforward module by 8 experts:



Resource savings:

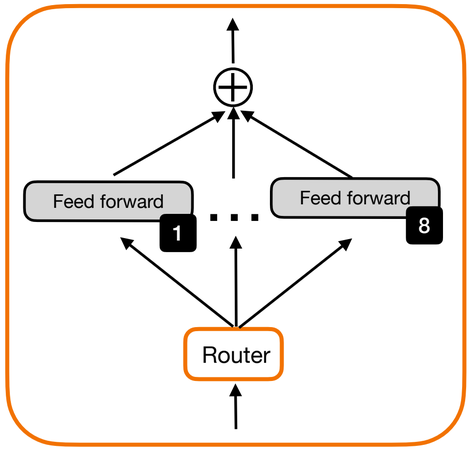
- Model size is 47B
- but only 2 experts are utilized at a time
- only 13B parameters are active at a time

Mixture of Experts (comeback)



Mixtral (Jan 2024)

replaces the feedforward module by 8 experts:

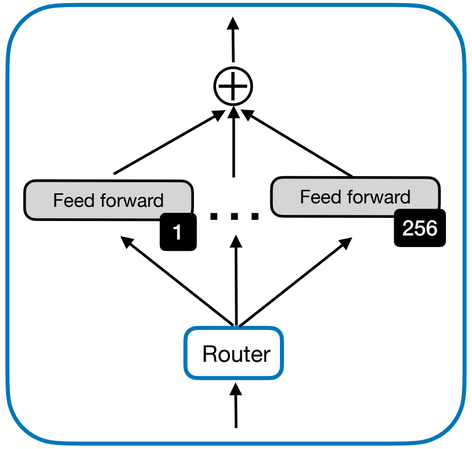


Resource savings:

- Model size is 47B
- but only 2 experts are utilized at a time
- only 13B parameters are active at a time

DeepSeek v3 (Dec 2024)

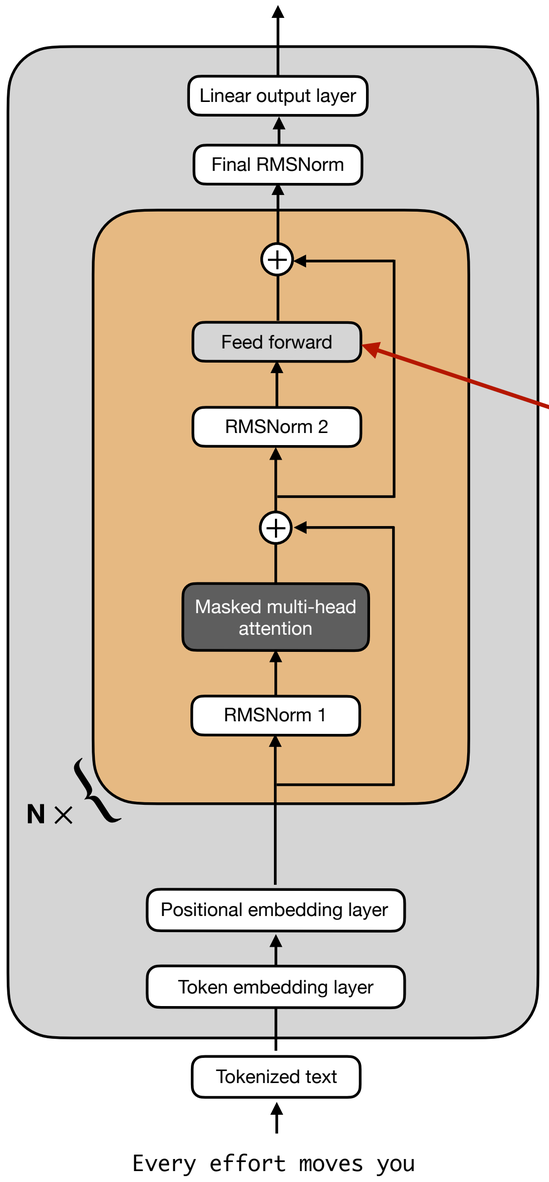
replaces the feedforward module by 256 experts:



Resource savings:

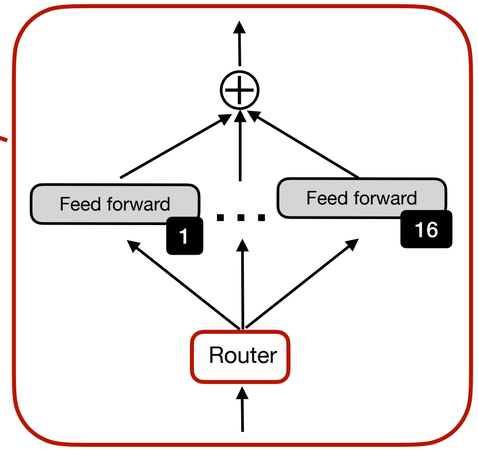
- Model size is 671B
- but only 8 experts are utilized at a time
- only 37B parameters are active at a time

Mixture of Experts (comeback)



Llama 4 Scout (Apr 2025)

replaces the feedforward module by 16 experts:

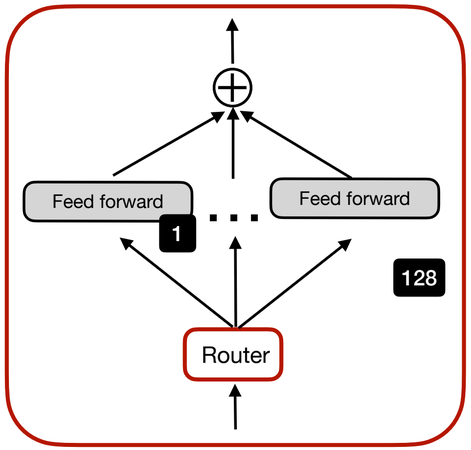


Resource savings:

- Model size is **109B**
- but only **2 experts?** are utilized at a time
- only **17B** parameters are active at a time

Llama 4 Maverick (Apr 2025)

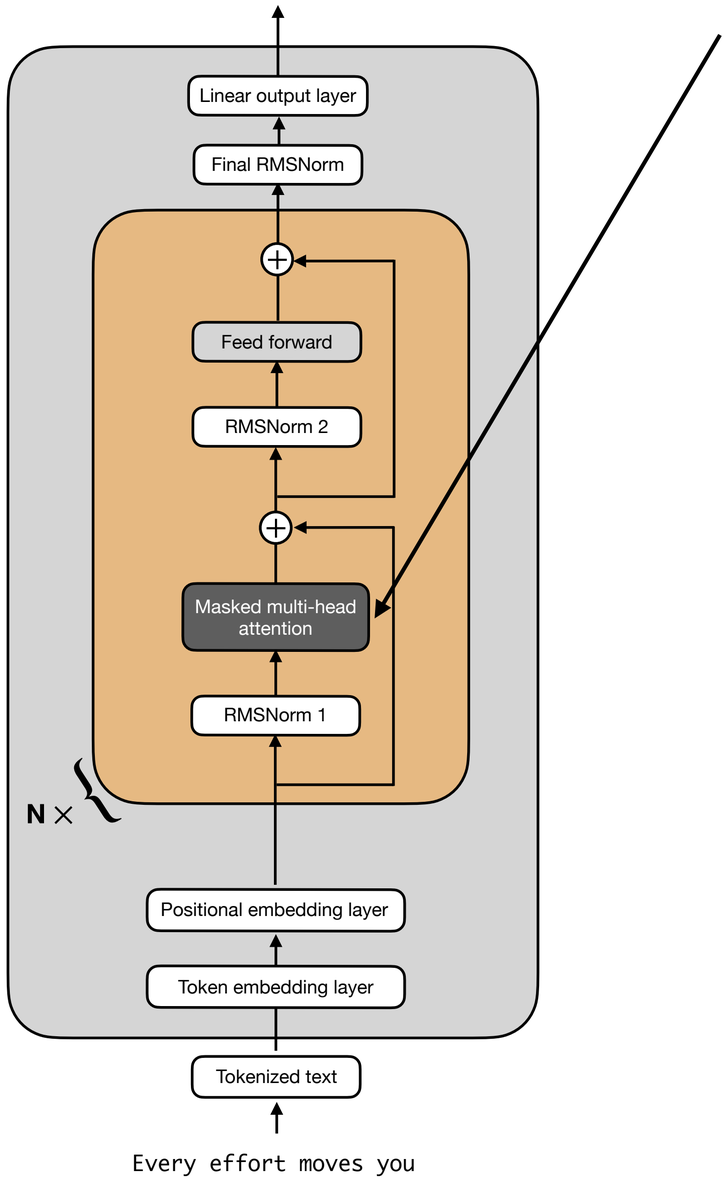
replaces the feedforward module by 128 experts:



Resource savings:

- Model size is **400B**
- but only **2 experts?** are utilized at a time
- only **17B** parameters are active at a time

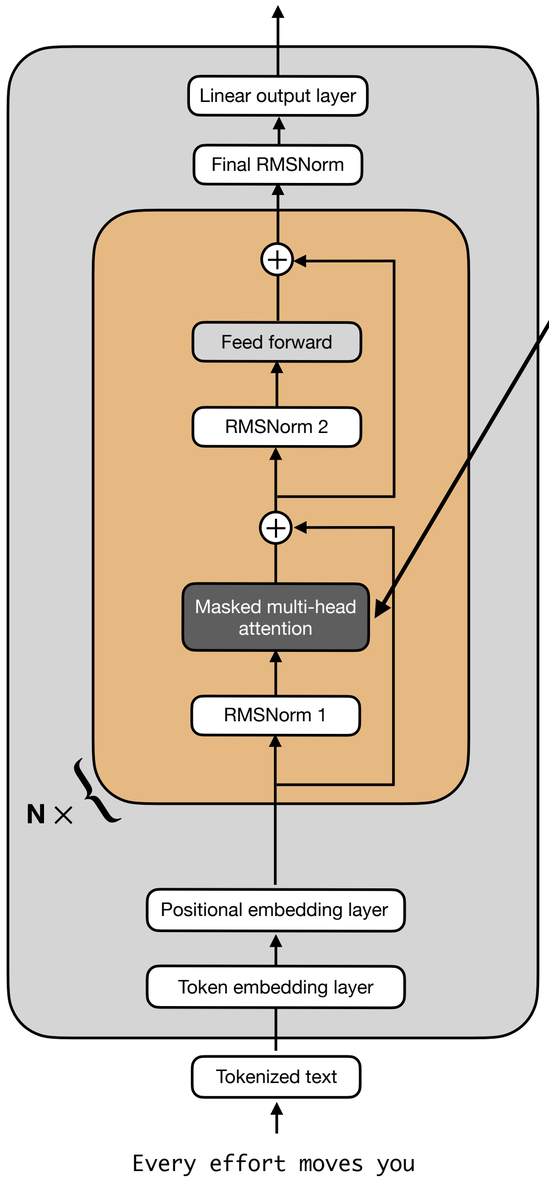
Multi-Head Latent Attention



Another trick to reduce memory usage

DeepSeek-V2 paper (May 2024)

Multi-Head Latent Attention

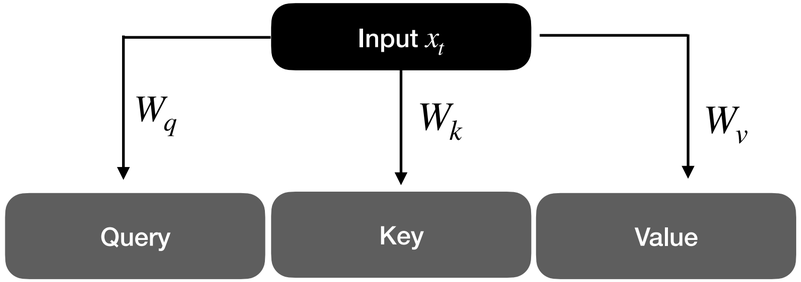


Another trick to reduce memory usage

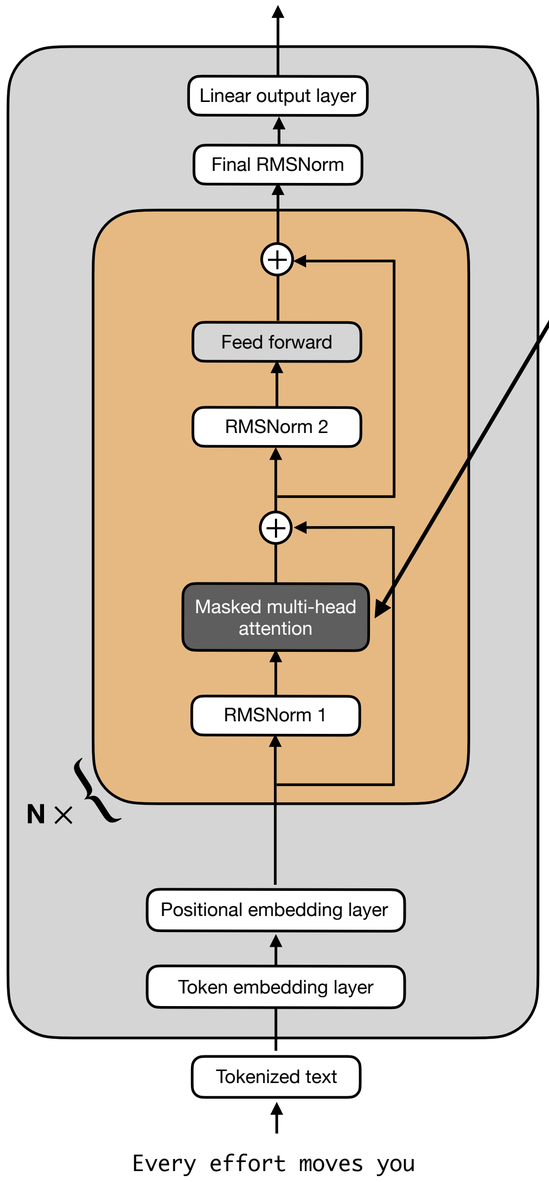
DeepSeek-V2 paper (May 2024)

Regular multi-head attention

Inference step *t:*



Multi-Head Latent Attention

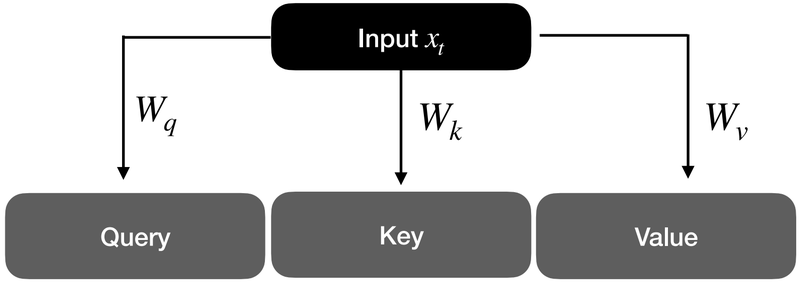


Another trick to reduce memory usage

DeepSeek-V2 paper (May 2024)

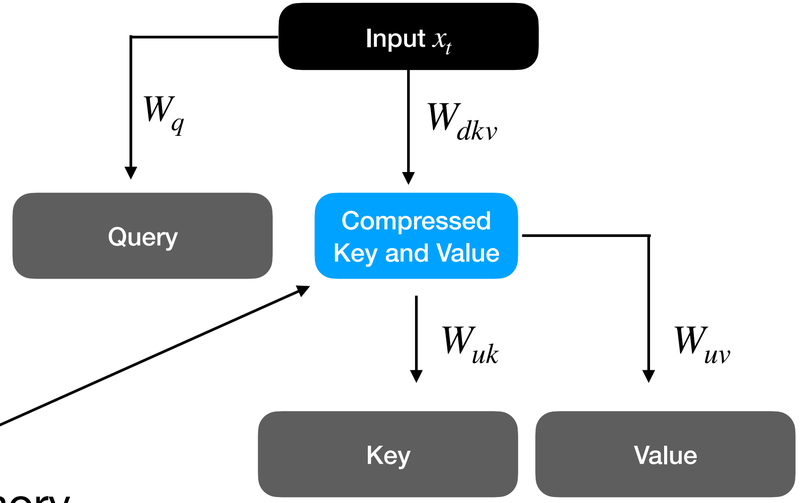
Regular multi-head attention

Inference step *t:*



Multi-head latent attention

Inference step *t:*



Key idea: Reduces memory usage in KV cache

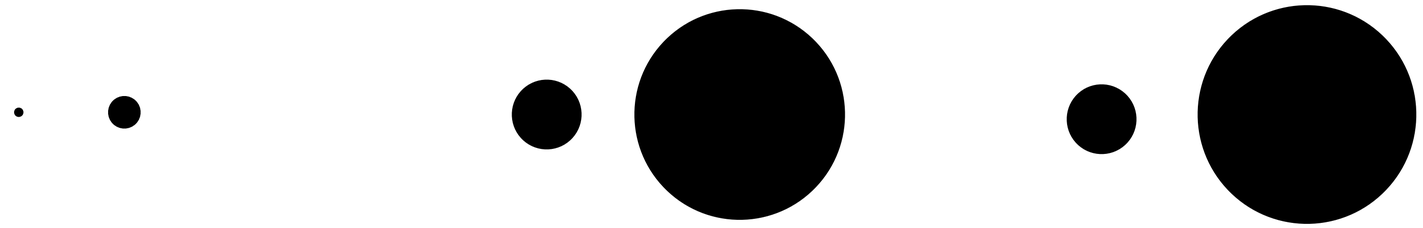
Model sizes over the years

2019	2023
GPT-2	Llama-1
124M to 1.5B	7B to 65B

Sebastian Raschka

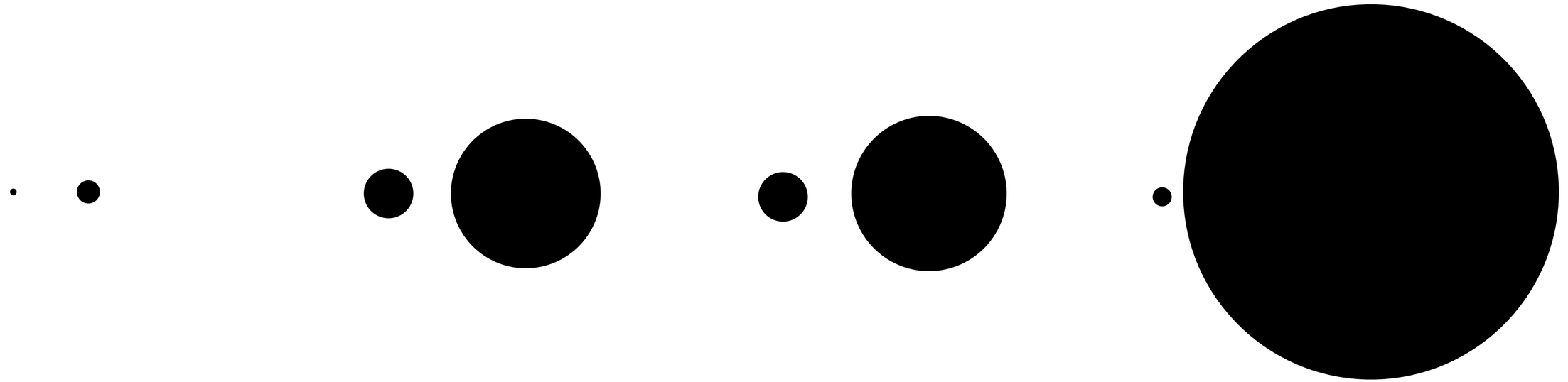
Model sizes over the years

2019	2023	2023
GPT-2	Llama-1	Llama-2
124M to 1.5B	7B to 65B	7B to 70B



Model sizes over the years

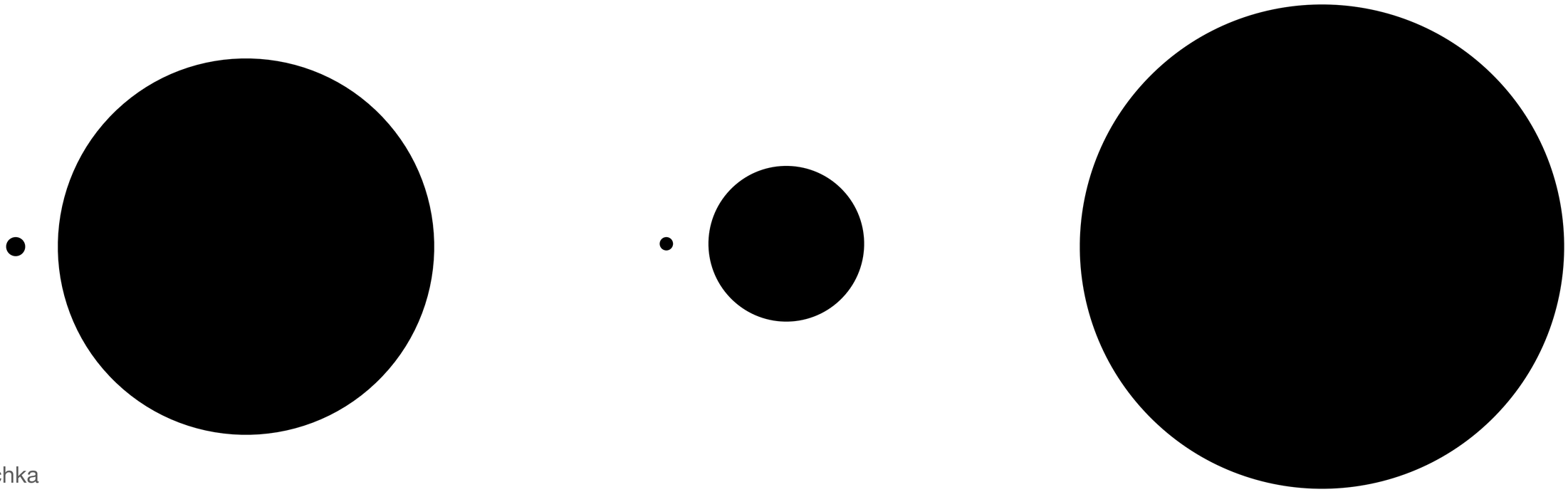
2019	2023	2023	2024
GPT-2	Llama-1	Llama-2	Llama-3
124M to 1.5B	7B to 65B	7B to 70B	1B to 405B



Sebastian Raschka

Model sizes over the years

Jul 2024	Sep 2024	Dec 2024
Llama-3	Qwen 2.5	DeepSeek v3 (MoE)
1B to 405B	0.5B to 72B	671B



Sebastian Raschka

Model sizes over the years

Mar 2025

Gemma-3

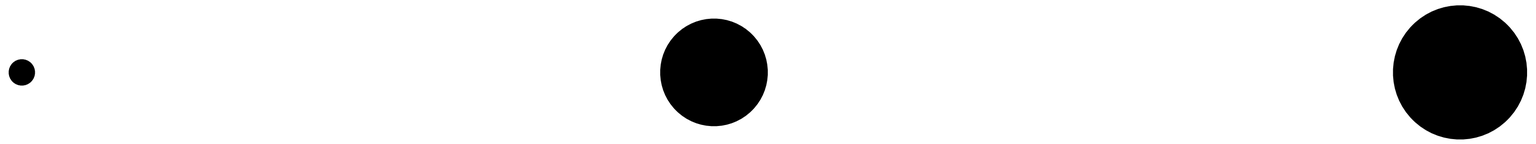
4B

Gemma-3

12B

Gemma-3

27B

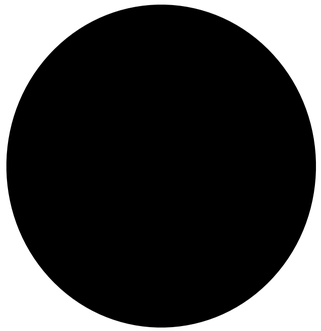


Model sizes over the years

April 2025

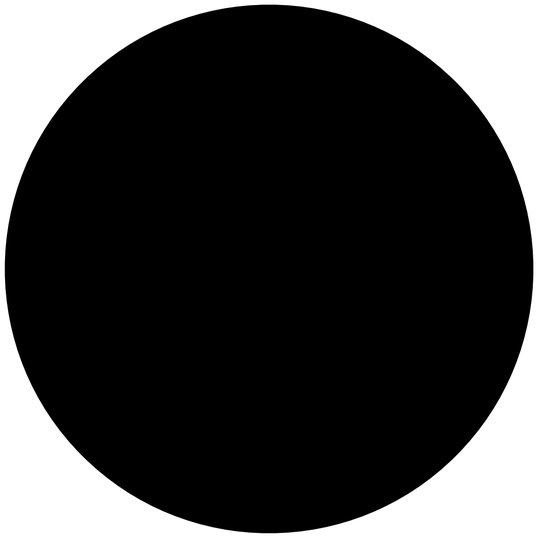
Llama 4 Scout
(MoE)

107B



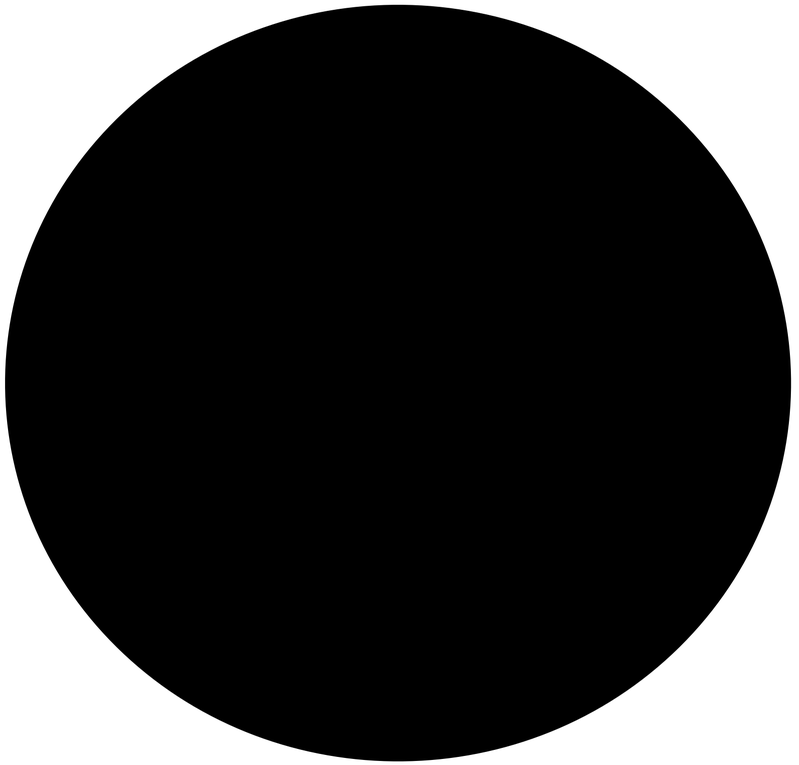
Llama 4 Maverick
(MoE)

400B

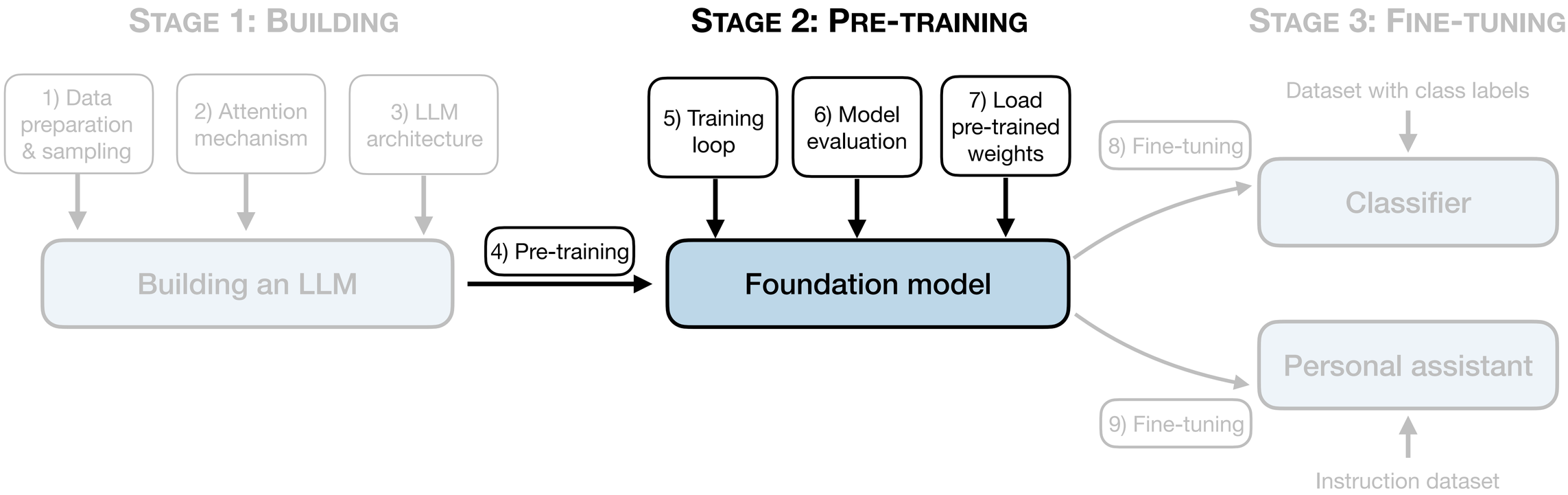


Llama 4 Behemoth
(MoE)

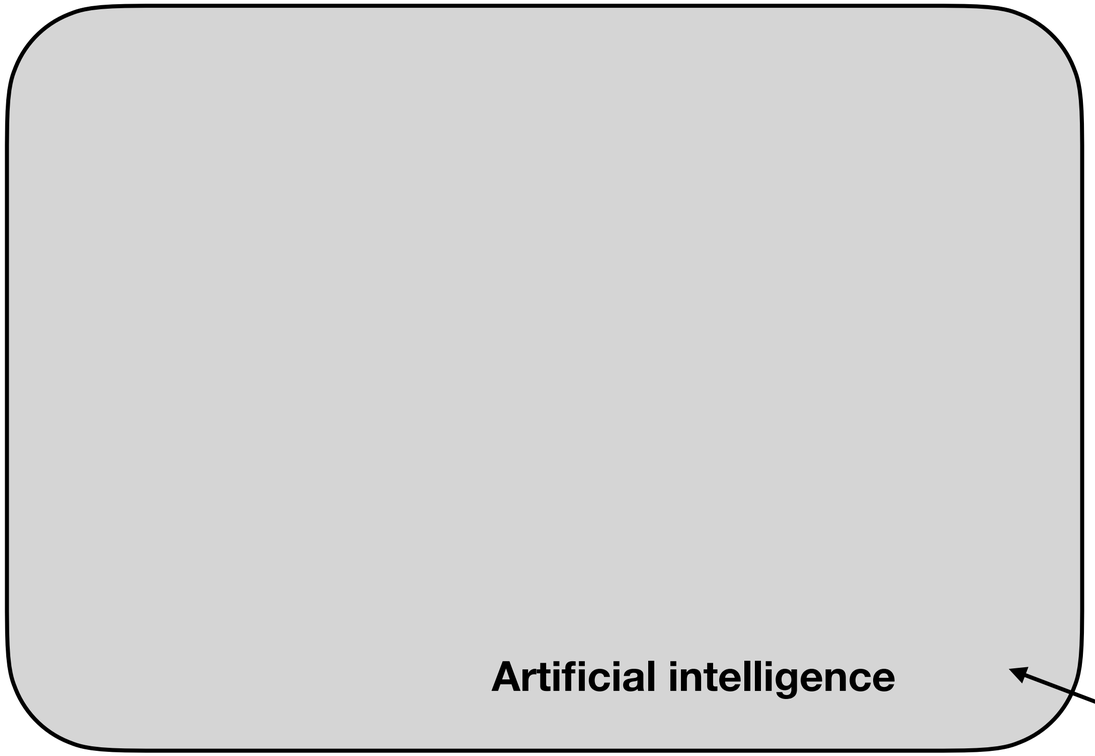
2T



Stage 2: Pre-training

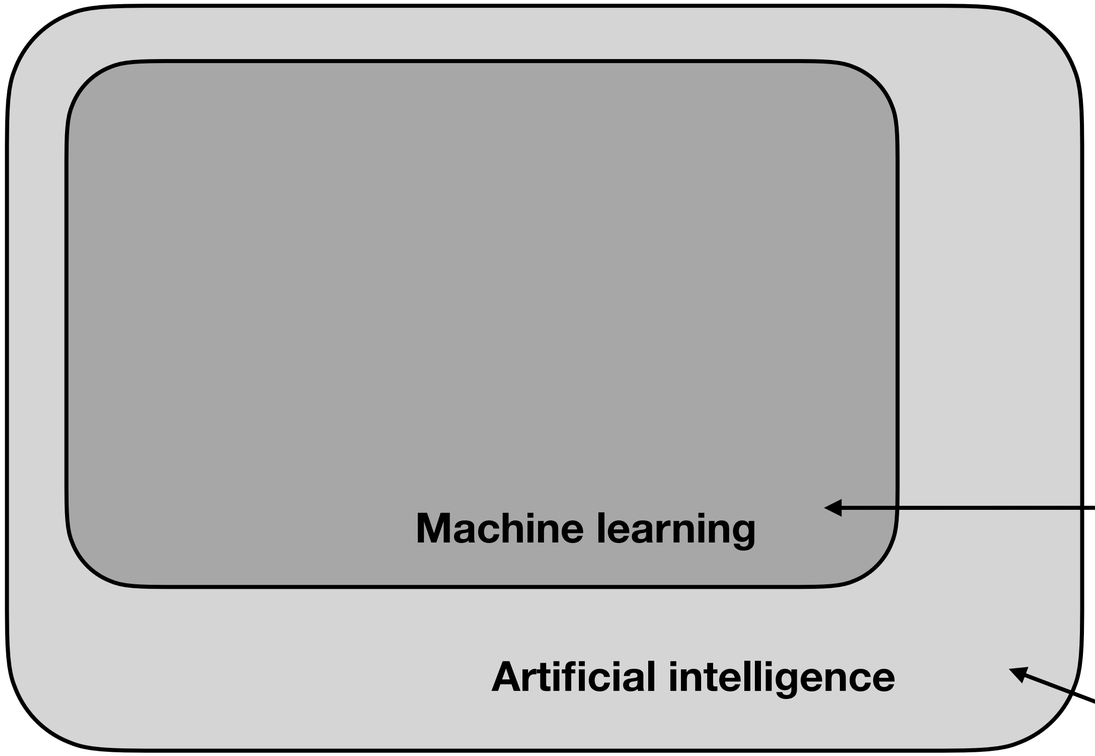

STAGE 1: BUILDING
STAGE 2: PRE-TRAINING
STAGE 3: FINE-TUNING
1) Data preparation & sampling
2) Attention mechanism
3) LLM architecture
Building an LLM
4) Pre-training
5) Training loop
6) Model evaluation
7) Load pre-trained weights
Foundation model
8) Fine-tuning
Dataset with class labels
Classifier
Personal assistant
9) Fine-tuning
Instruction dataset

LLMs are deep neural networks



Automated systems for decision making (with ambition to mimic human-like intelligence)

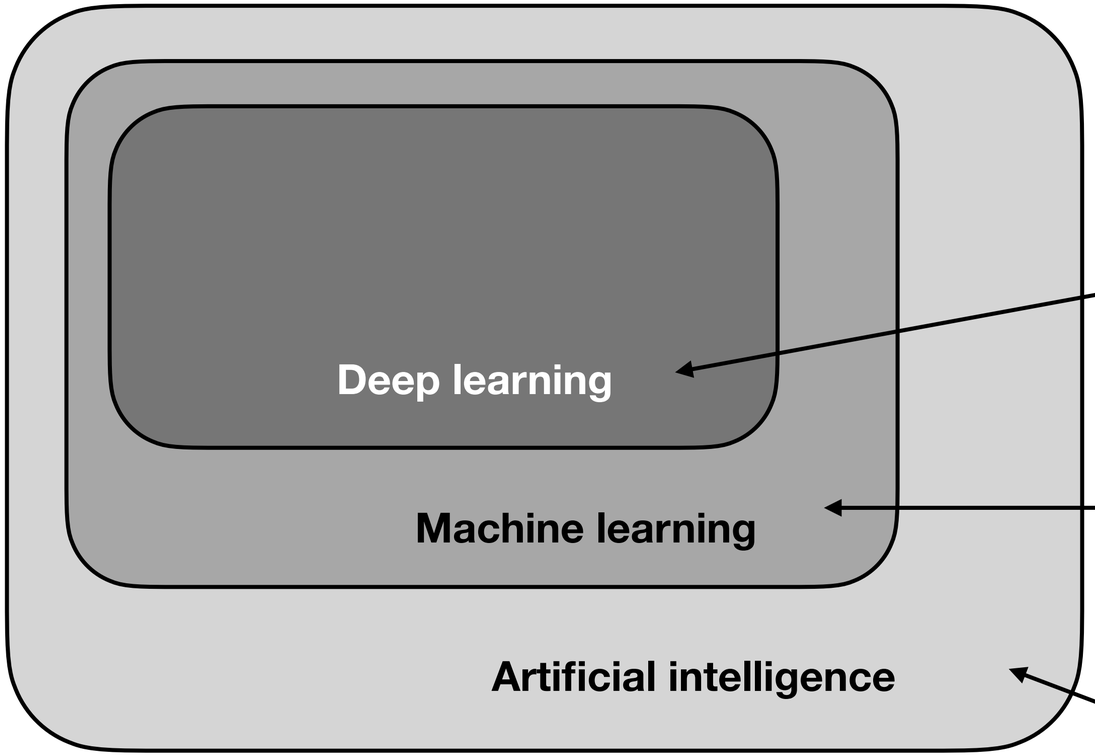
LLMs are deep neural networks



Algorithms that learn rules automatically from data

Automated systems for decision making (with ambition to mimic human-like intelligence)

LLMs are deep neural networks

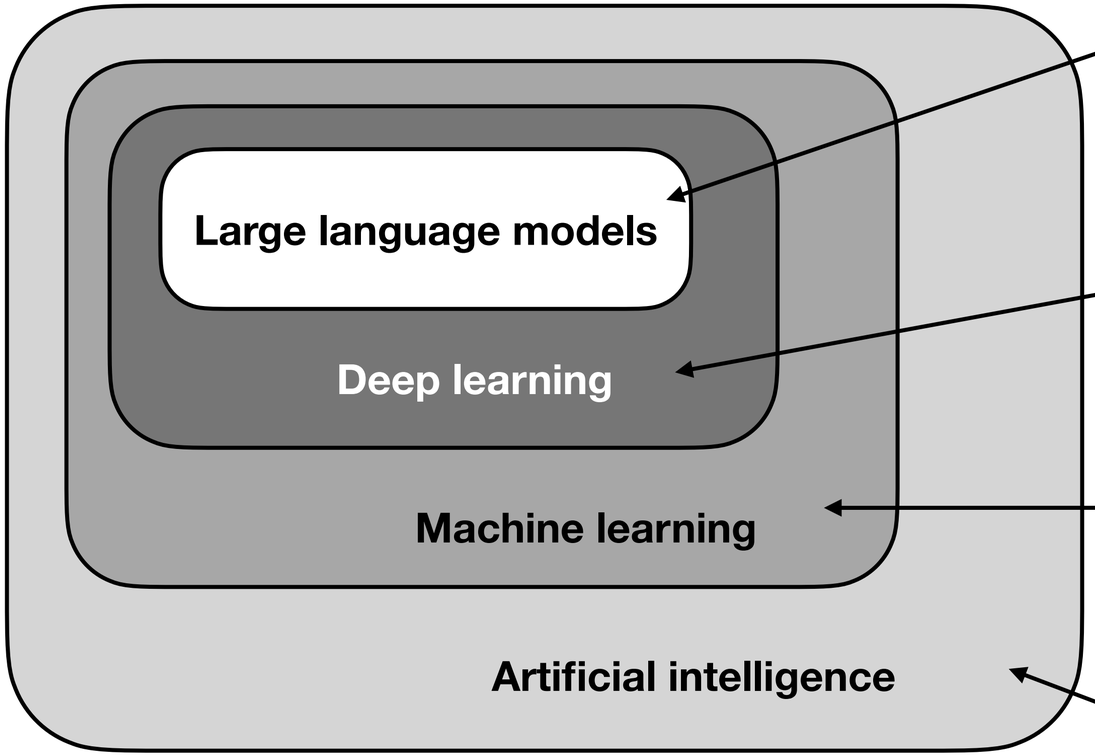


Machine learning with neural networks consisting of many layers

Algorithms that learn rules automatically from data

Automated systems for decision making (with ambition to mimic human-like intelligence)

LLMs are deep neural networks



Deep neural network for parsing and generating human-like text

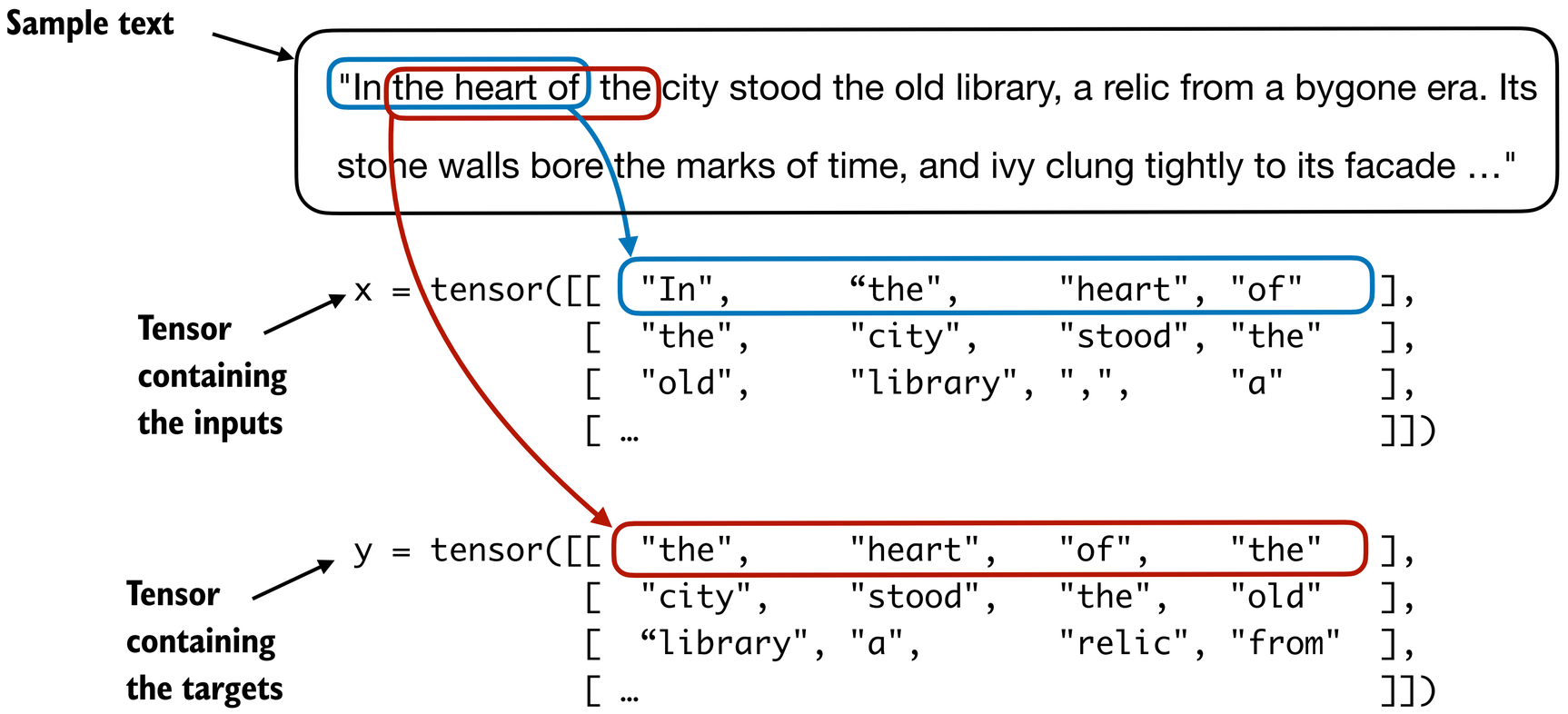
Machine learning with neural networks consisting of many layers

Algorithms that learn rules automatically from data

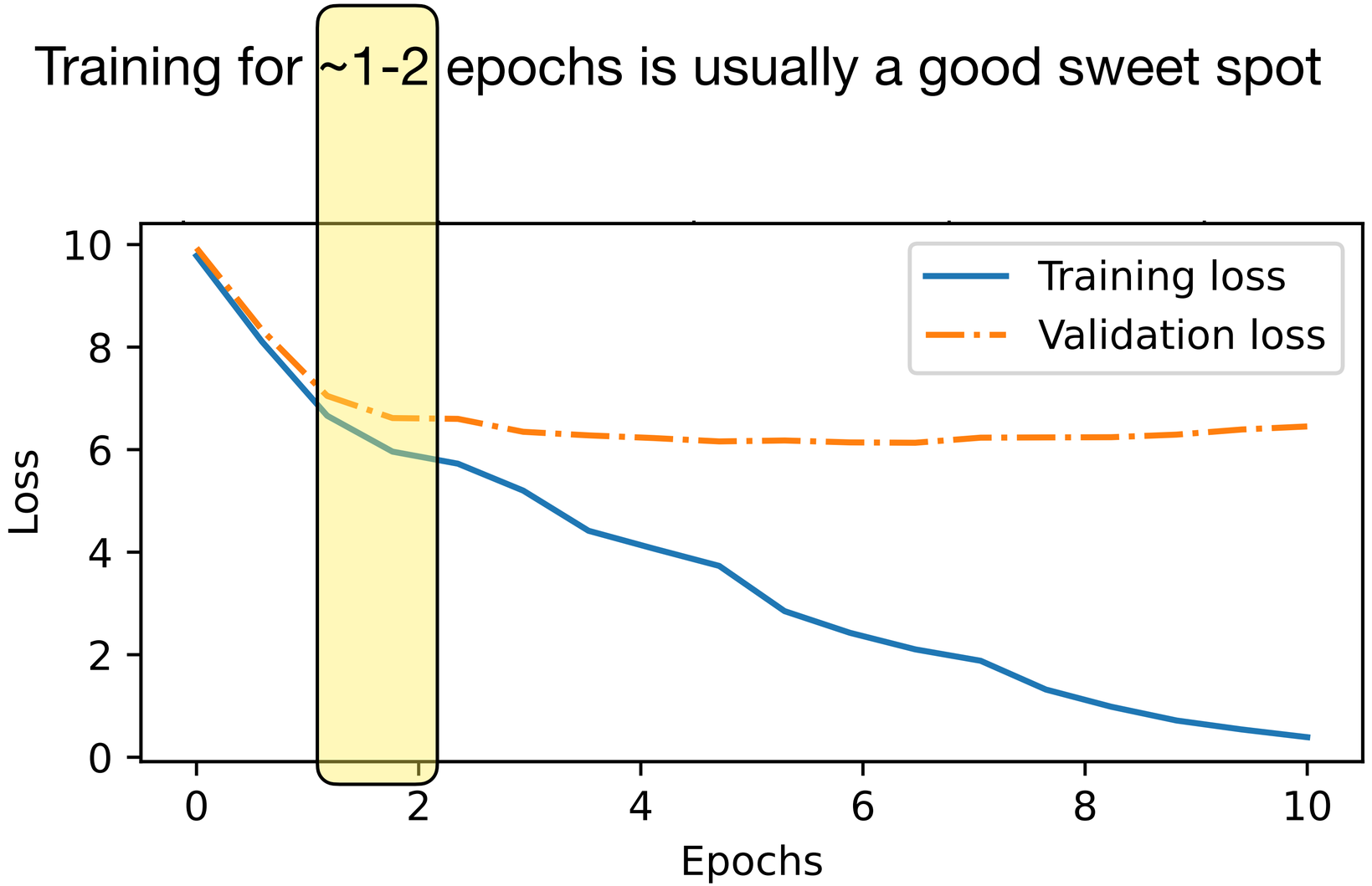
Automated systems for decision making (with ambition to mimic human-like intelligence)

Pretty standard deep learning training loop

Labels are the inputs shifted by +1



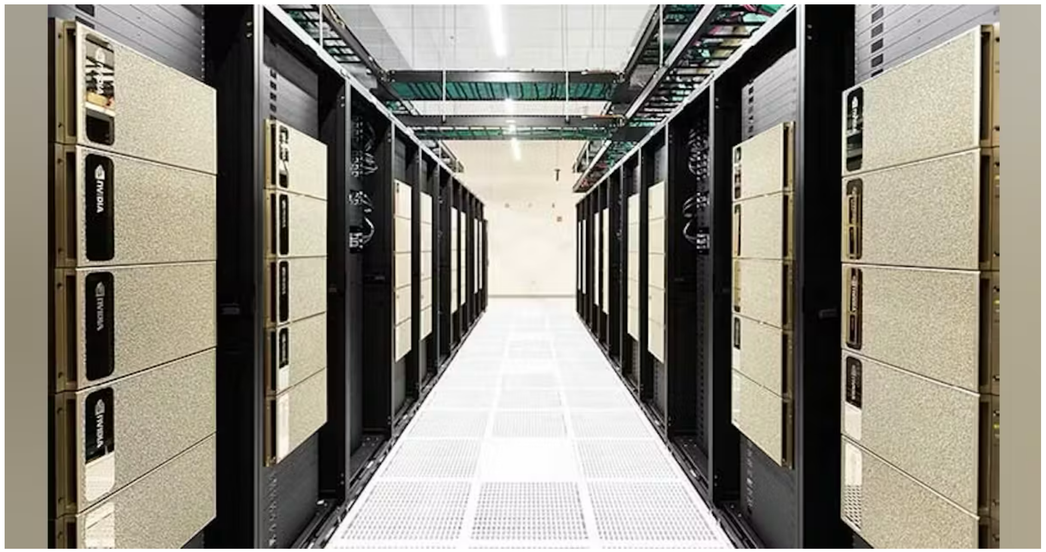
Training for ~1-2 epochs is usually a good sweet spot



“Pretty standard” deep learning training loop

~2018

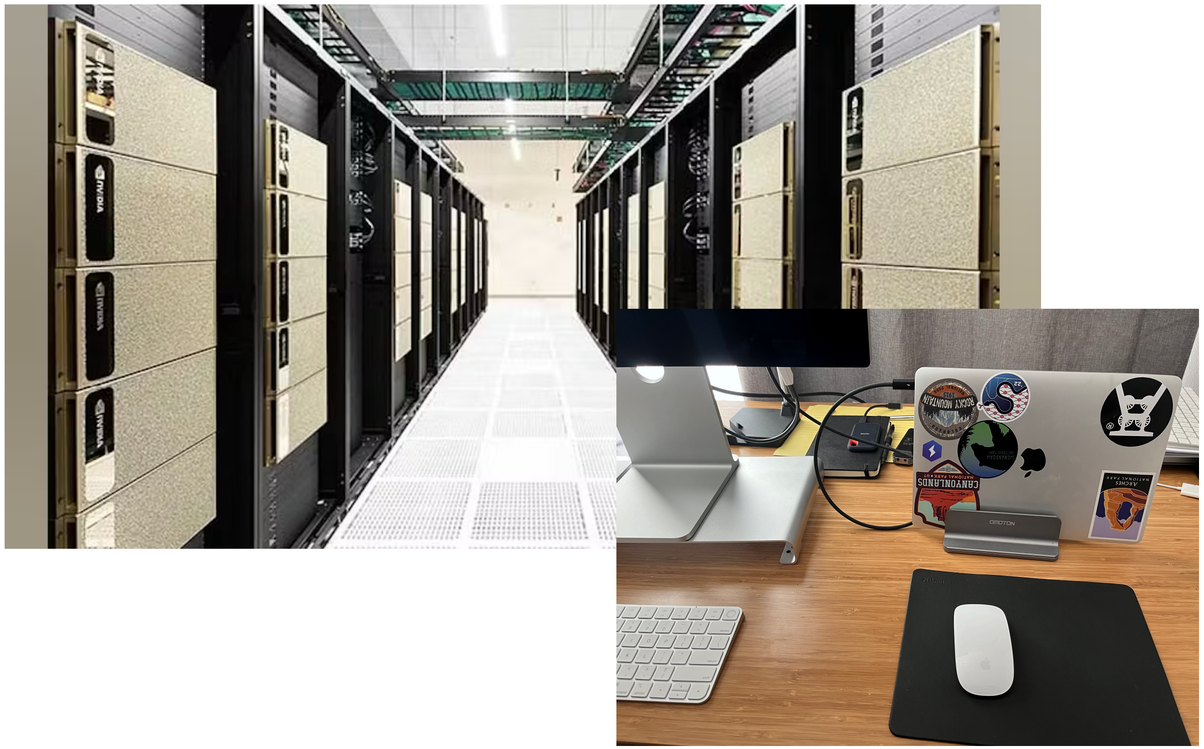
Today



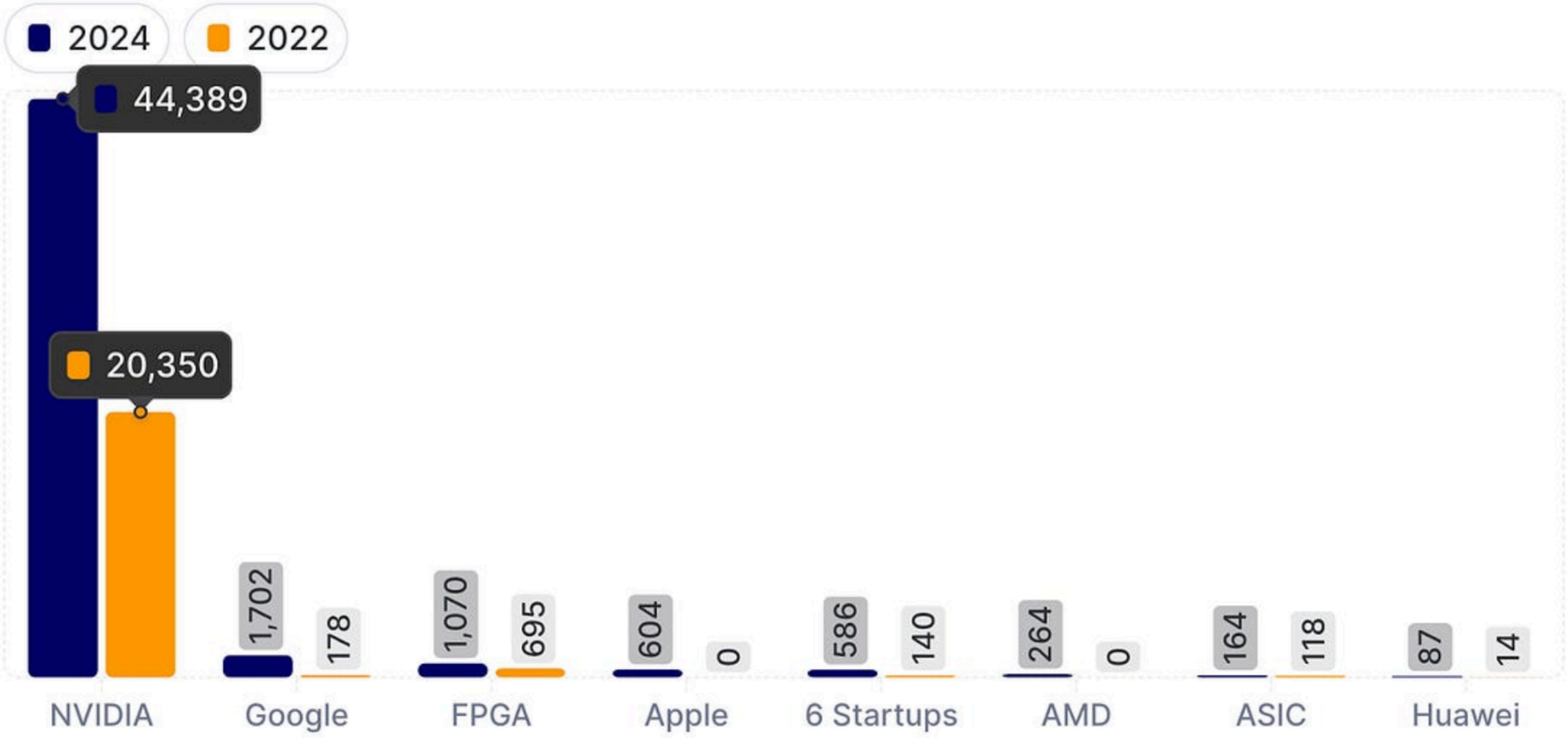
“Pretty standard” deep learning training loop

~2018

Today

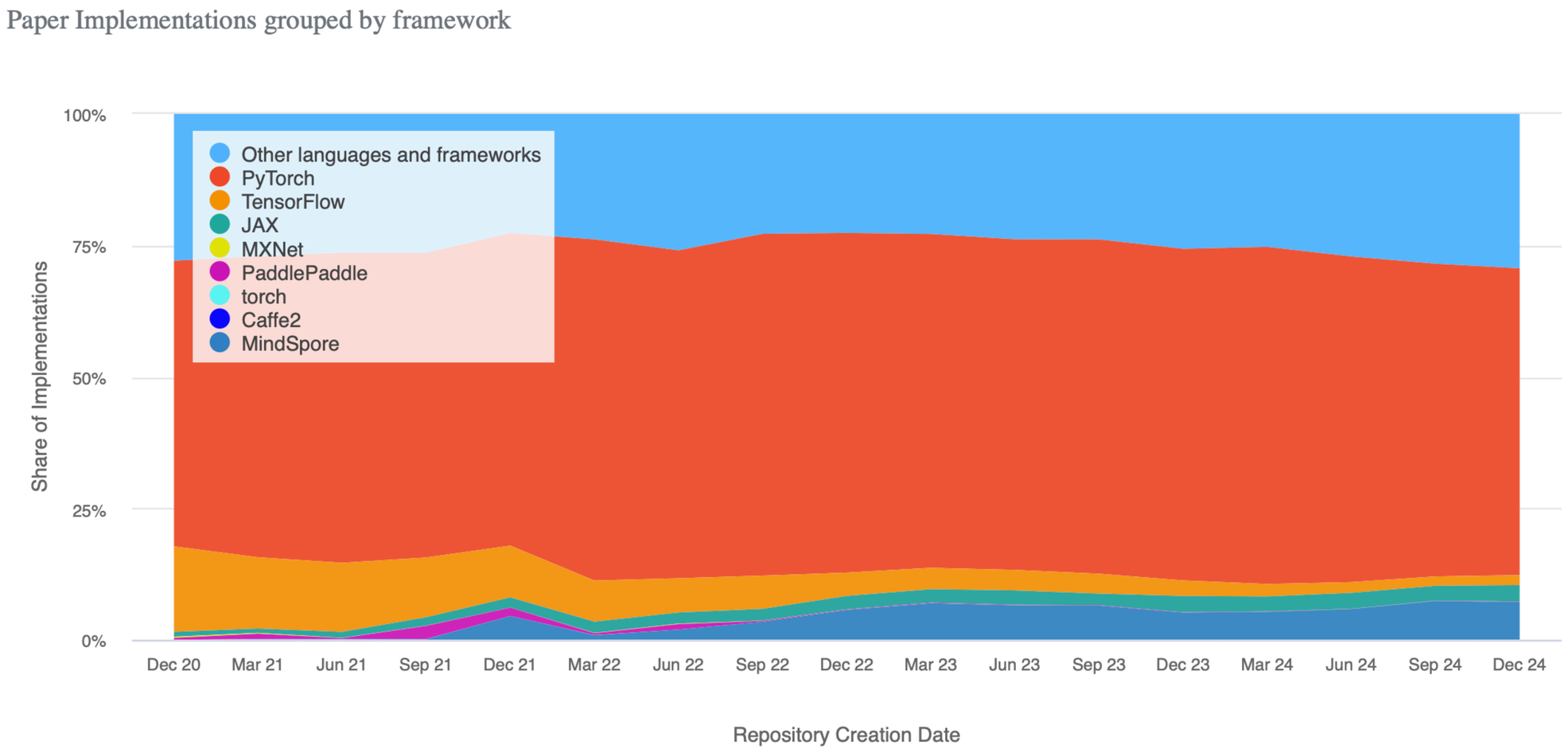


Most AI Research Runs on NVIDIA Hardware



Source: https://press.airstreet.com/p/guide-to-ai-january-2025

Most AI research happens with PyTorch



Source: https://paperswithcode.com/trends

Audience question

If training LLMs is “just” training deep neural nets…

What makes it so hard / inaccessible?

Complexity comes from …

1. Being able to afford
2. Getting access to
3. Managing

… compute resources

Training Costs	Pre-Training	Context Extension	Post-Training	Total
in H800 GPU Hours	2664K	119K	5K	2788K
in USD	$5.328M	$0.238M	$0.01M	$5.576M

Table 1 | Training costs of DeepSeek-V3, assuming the rental price of H800 is $2 per GPU hour.

DeepSeek-V3 Technical Report (27 Dec, 2024) https://arxiv.org/abs/2412.19437

Metric	DeepSeek V3	Llama 3.1
Parameters	671B total (37B active per token)	405B
GPU Type	NVIDIA H800	NVIDIA H100
GPU Count	2,048	Up to 16,000
Training Duration	~2 months	~2.6 months (estimated)
Tokens Processed	14.8T	15.6T
GPU Hours	2.788M	~30.8M
Training Cost	~$5.6M	~$92.4M–123.2M (estimated)
Cost per Trillion Tokens	~$378K	~$5.93M–7.90M

Note: Cost estimations uses an average of **$2/hour** for H800 GPUs (DeepSeek V3) and **$3/hour** for H100 GPUs (Llama 3.1) based on rental GPU prices.

Source: https://apxml.com/posts/training-cost-deepseek-v3-vs-llama-3

3.3.4 Reliability and Operational Challenges

The complexity and potential failure scenarios of 16K GPU training surpass those of much larger CPU clusters that we have operated. Moreover, the synchronous nature of training makes it less fault-tolerant—a single GPU failure may require a restart of the entire job. Despite these challenges, for Llama 3, we achieved higher than 90% effective training time while supporting automated cluster maintenance, such as firmware and Linux kernel upgrades (Vigraham and Leonhardi, 2024), which resulted in at least one training interruption daily. The effective training time measures the time spent on useful training over the elapsed time.

During a 54-day snapshot period of pre-training, we experienced a total of 466 job interruptions. Of these, 47 were planned interruptions due to automated maintenance operations such as firmware upgrades or operator-initiated operations like configuration or dataset updates. The remaining 419 were unexpected interruptions, which are classified in Table 5. Approximately 78% of the unexpected interruptions are attributed to confirmed hardware issues, such as GPU or host component failures, or suspected hardware-related issues like silent data corruption and unplanned individual host maintenance events. GPU issues are the largest category, accounting for 58.7% of all unexpected issues. Despite the large number of failures, significant manual intervention was required only three times during this period, with the rest of issues handled by automation.

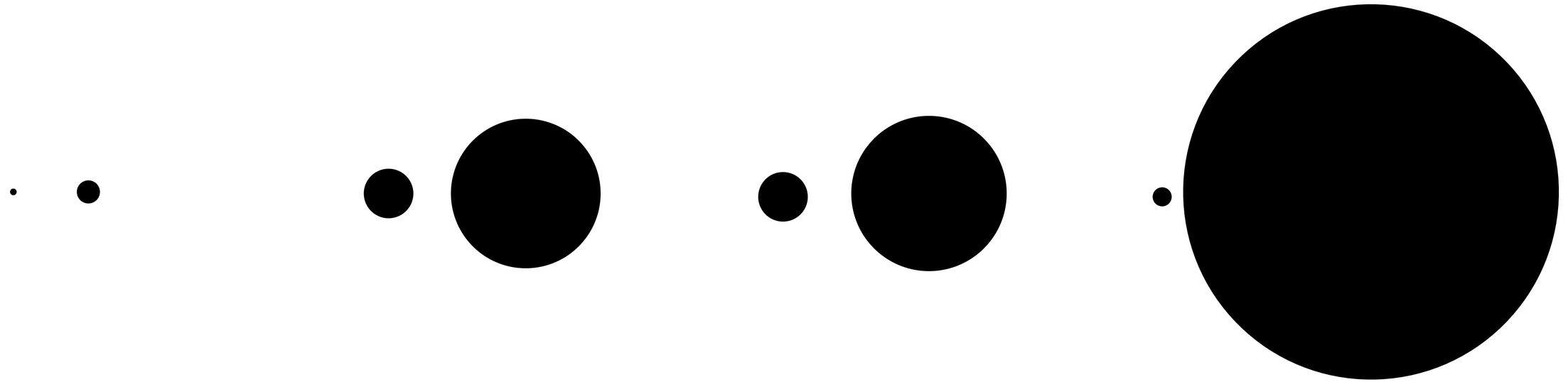
The Llama 3 Herd of Models (31 Jul 2024), https://arxiv.org/abs/2407.21783

Increasing price and complexity due to

- larger models
- larger datasets
- multi-stage training

Model sizes over the years

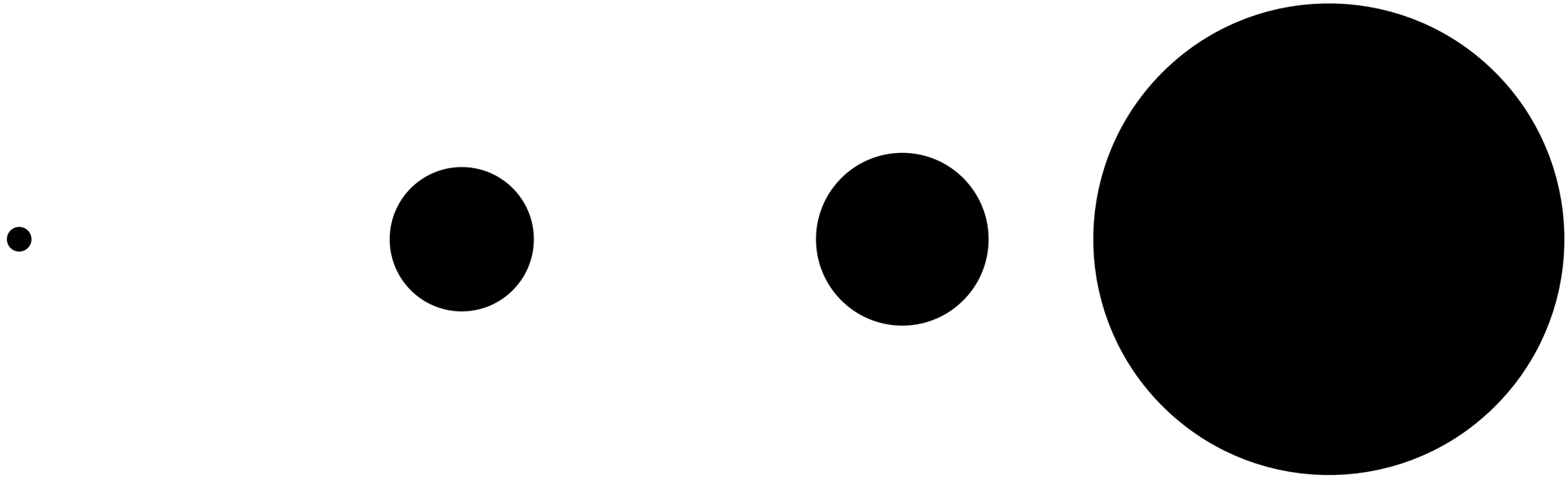
2019	2023	2023	2024
GPT-2	Llama-1	Llama-2	Llama-3
124M to 1.5B	7B to 65B	7B to 70B	1B to 405B



Sebastian Raschka

Pre-training **dataset** sizes over the years

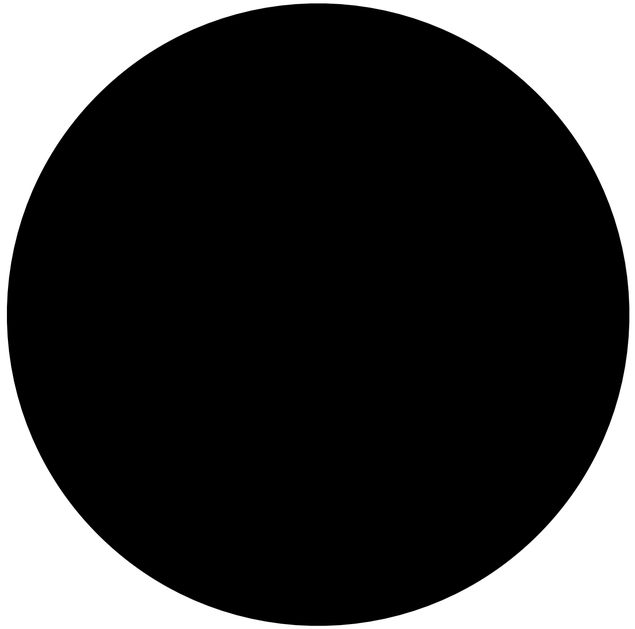
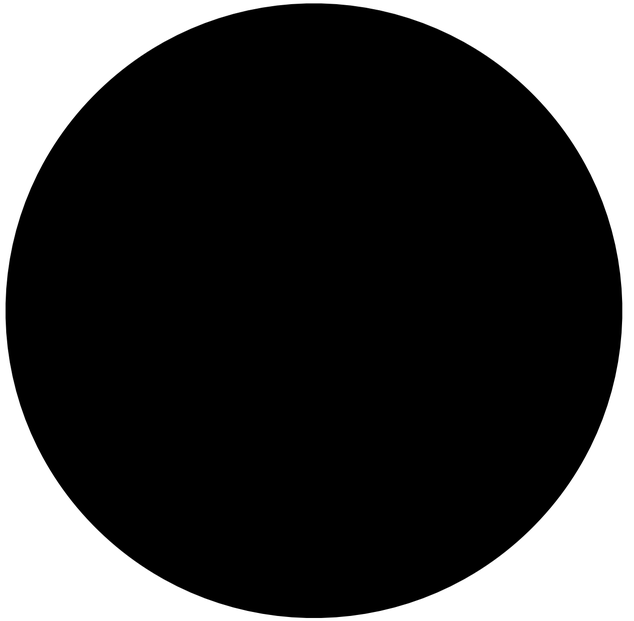
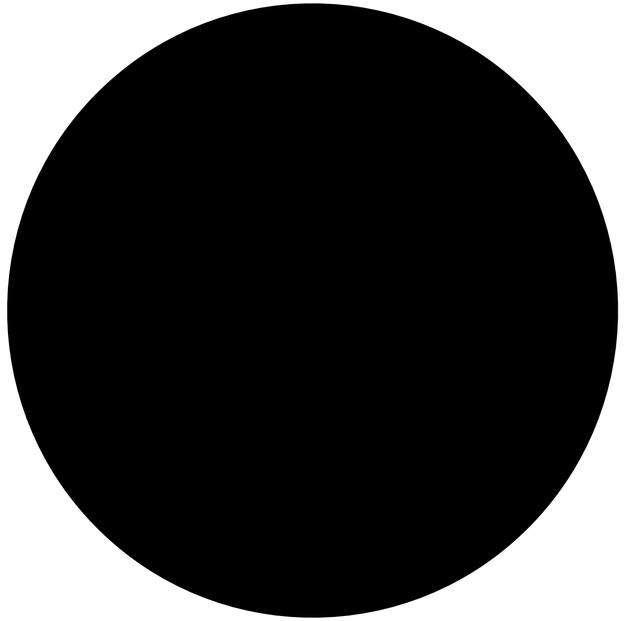
2019	2023	2023	2024
GPT-2	Llama-1	Llama-2	Llama-3
40B tokens	1.4T tokens	2T tokens	15T tokens



Sebastian Raschka

Pre-training **dataset** sizes over the years

Jul 2024	Dec 2024	Mar 2025
Llama-3	DeepSeek-V3	Gemma 3
15T tokens	14.8T tokens	14T tokens

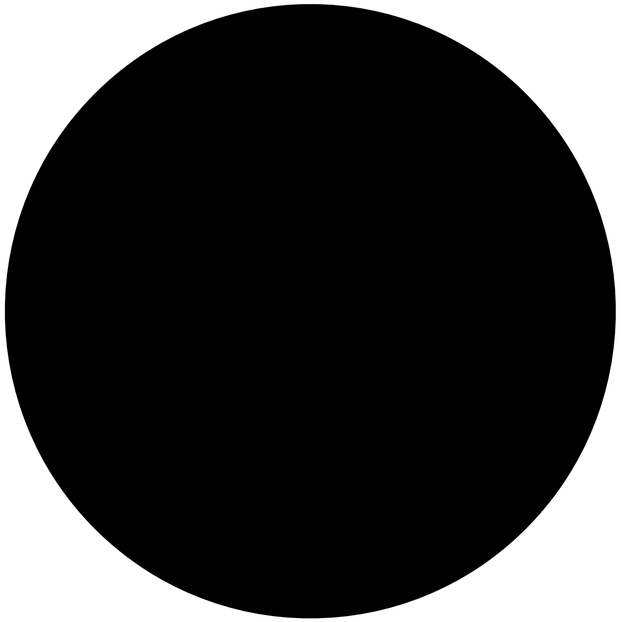




Pre-training **dataset** sizes over the years

Mar 2025

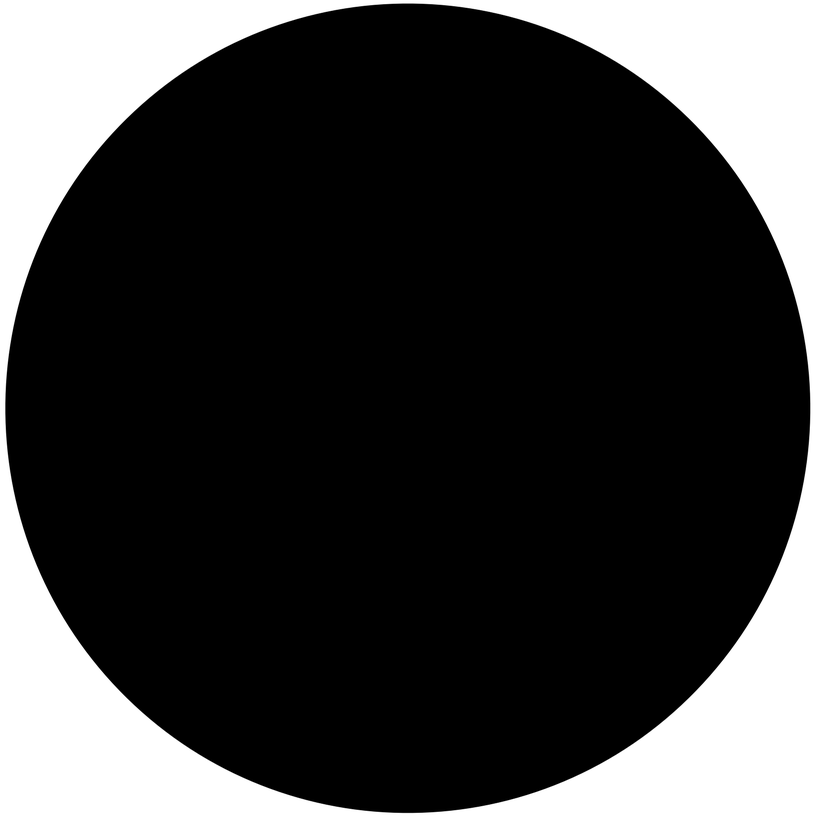
Gemma 3

14T tokens



Apr 2025

Llama 4 30T tokens



What’s new dataset-wise

- **Data filtering** to remove low-quality data

What's new dataset-wise

- **Data filtering** to remove low-quality data
- Enhancing **data mixing** to increase data diversity

What's new dataset-wise

- **Data filtering** to remove low-quality data
- Enhancing **data mixing** to increase data diversity
- **Synthesize** additional pretraining data

What's new dataset-wise

- **Data filtering** to remove low-quality data
- Enhancing **data mixing** to increase data diversity
- **Synthesize** additional pretraining data
- Include **Q&A** (instruction) data in pretraining (Qwen, Apple FM)

What's new dataset-wise

- **Data filtering** to remove low-quality data
- Enhancing **data mixing** to increase data diversity
- **Synthesize** additional pretraining data
- Include **Q&A** (instruction) data in pretraining (Qwen, Apple FM)
- **Multi-stage** pretraining (long context, high-quality annealing)

Pre-training in 2024

Llama 3.1

- ✓ Filtering
- ✓ Synthetic data
- ✓ Mixing
- ✗ Q&A format
- ✓ Long-context stage
- ✓ Continued pre-training
- ✓ High-quality stage
- ✗ Knowledge distillation

Qwen 2

- ✓ Filtering
- ✓ Synthetic data
- ✓ Mixing
- ✓ Q&A format
- ✓ Long-context stage
- ✓ Continued pre-training
- ✓ High-quality stage
- ✗ Knowledge distillation

Apple FM

- ✓ Filtering
- ✓ Synthetic data
- ✓ Mixing
- ✓ Q&A format
- ✓ Long-context stage
- ✓ Continued pre-training
- ✓ High-quality stage
- ✓ Knowledge distillation

Gemma 2

- ✓ Filtering
- ✗ Synthetic data
- ✓ Mixing
- ✗ Q&A format
- ✗ Long-context stage
- ✗ Continued pre-training
- ✗ High-quality stage
- ✓ Knowledge distillation

The Llama 3 Herd of Models (31 July 2024), https://arxiv.org/abs/2407.21783

Qwen2 Technical Report (15 July 2024), https://arxiv.org/abs/2407.10671

Apple Intelligence Foundation Language Models (29 July 2024), https://arxiv.org/abs/2407.21075

Gemma 2: Improving Open Language Models at a Practical Size (31 July 2024), https://arxiv.org/abs/2408.0011

New LLM Pre-training and Post-training Paradigms

A Look at How Modern LLMs Are Trained

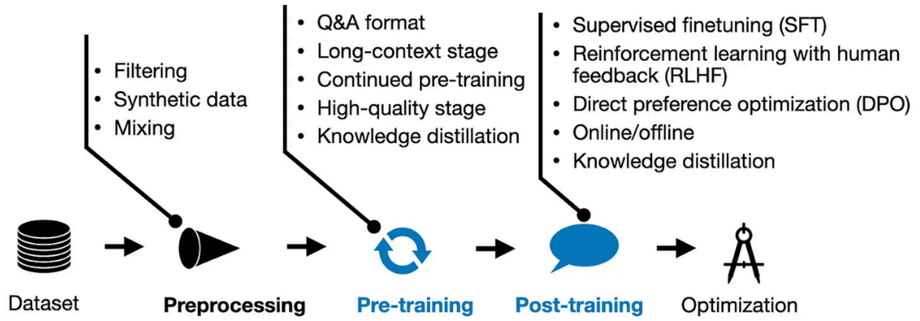
SEBASTIAN RASCHKA, PHD

AUG 17, 2024

257 23 25 Share

The development of large language models (LLMs) has come a long way, from the early GPT models to the sophisticated open-weight LLMs we have today. Initially, the LLM training process focused solely on pre-training, but it has since expanded to include both pre-training and post-training. Post-training typically encompasses supervised instruction fine-tuning and alignment, which was popularized by ChatGPT.

Training methodologies have evolved since ChatGPT was first released. In this article, I review the latest advancements in both pre-training and post-training methodologies, particularly those made in recent months.



An overview of the LLM development and training pipeline, with a focus on new pre-training and post-training methodologies discussed in this article

https://magazine.sebastianraschka.com/p/new-llm-pre-training-and-post-training

Audience question

Should you pre-train models?

Tip: Don’t pre-train models!

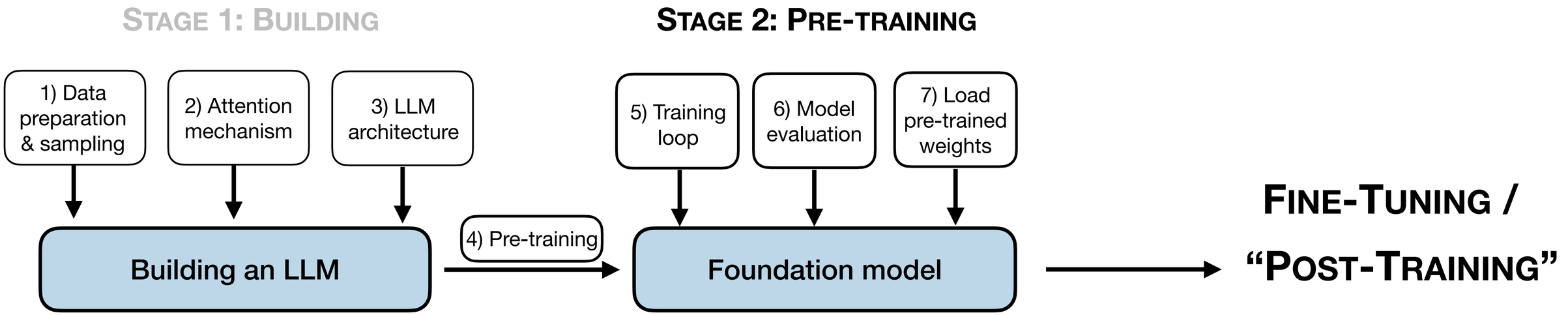
- Many open-weight models (0.5 B to 672 B) are available
- Focus on “fine-tuning” / “post-training”

Training Costs	Pre-Training	Context Extension	Post-Training	Total
in H800 GPU Hours	2664K	119K	5K	2788K
in USD	$5.328M	$0.238M	$0.01M	$5.576M

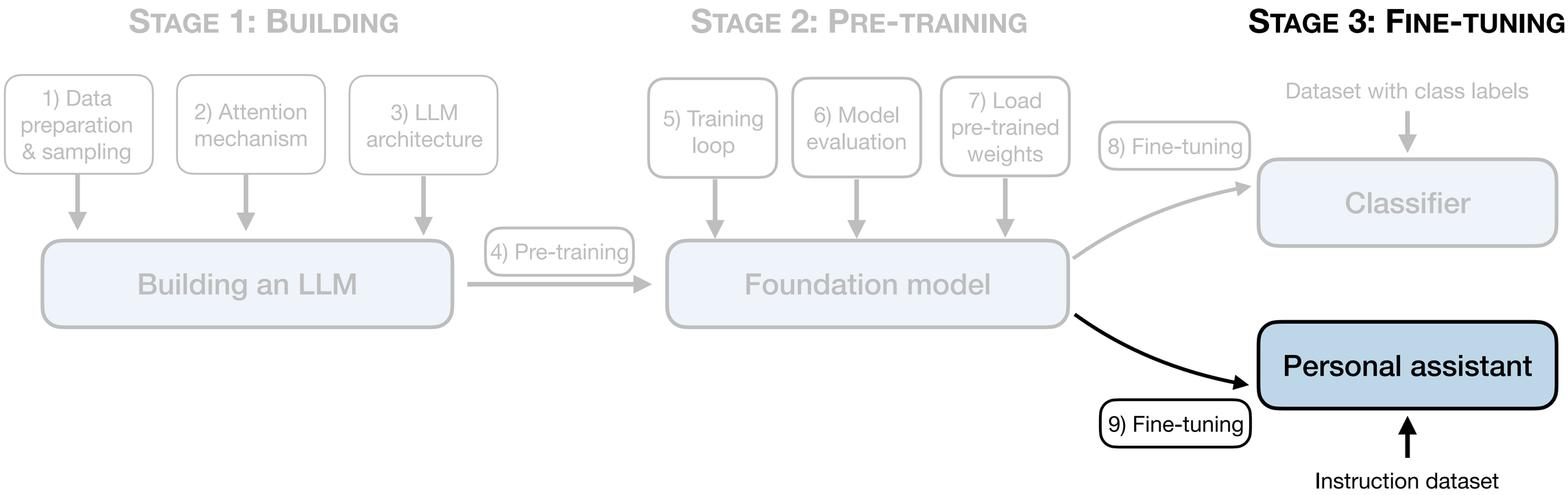
Table 1 | Training costs of DeepSeek-V3, assuming the rental price of H800 is $2 per GPU hour.

DeepSeek-V3 Technical Report (27 Dec, 2024) https://arxiv.org/abs/2412.19437

Stage 3: Fine-tuning / Post-Training


STAGE 1: BUILDING
STAGE 2: PRE-TRAINING
1) Data preparation & sampling
2) Attention mechanism
3) LLM architecture
Building an LLM
4) Pre-training
5) Training loop
6) Model evaluation
7) Load pre-trained weights
Foundation model
FINE-TUNING / "POST-TRAINING"

Instruction fine-tuning



Instruction fine-tuning

Also known as “supervised fine-tuning” or “SFT”

Instruction finetuning datasets

```
{
    "instruction": "Rewrite the following sentence
                    using passive voice.",
    "input": "The team achieved great results.",
    "output": "Great results were achieved by the
               team."
},
```

```
{
    "instruction": "Rewrite the following sentence using passive voice.",
    "input": "The team achieved great results.",
    "output": "Great results were achieved by the team."
},
```

Apply prompt style template (for example, Alpaca-style)

```
Below is an instruction that describes a task. Write a response
that appropriately completes the request.

### Instruction:
Rewrite the following sentence using passive voice.

### Input:
The team achieved great results.

### Response:
Great results were achieved by the team.
```

Pass to LLM for supervised instruction finetuning

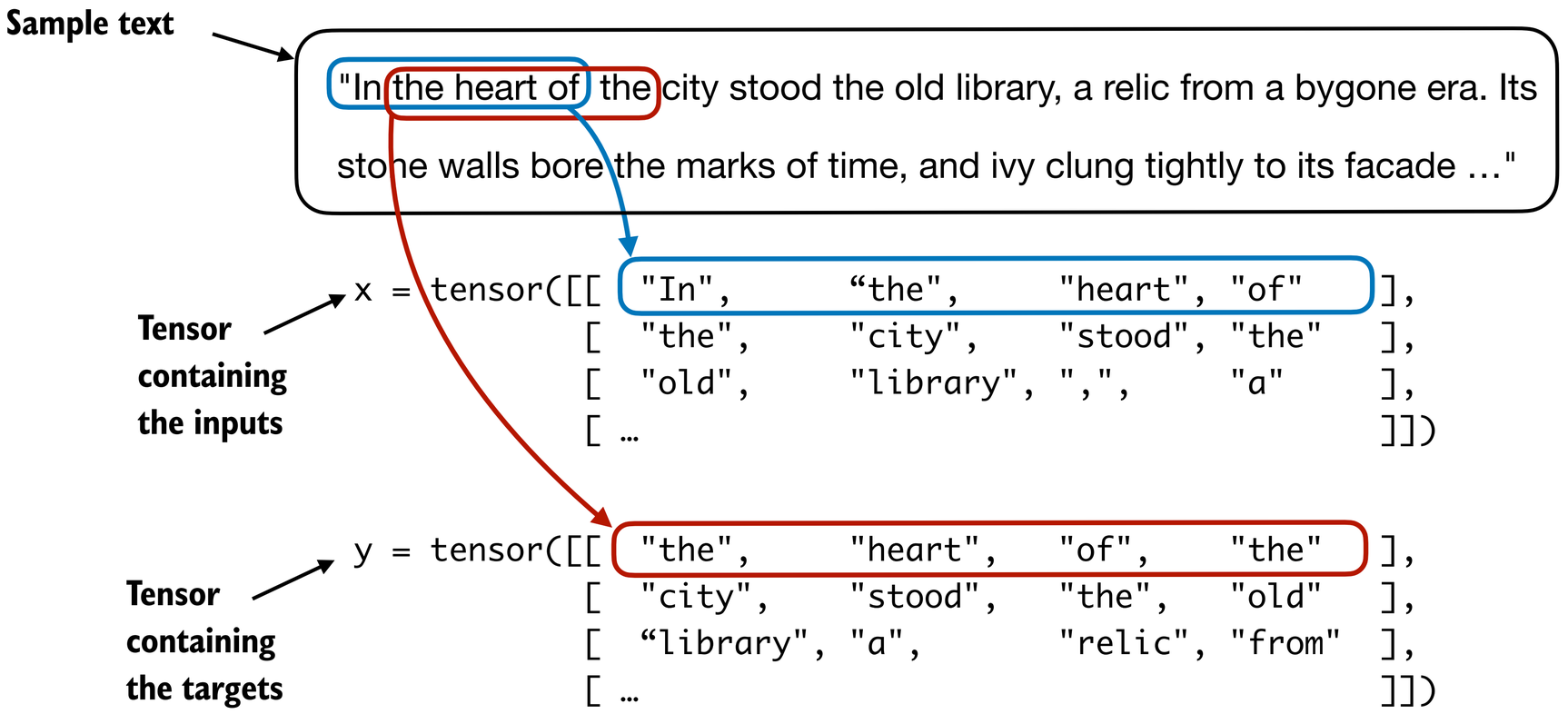
But model only sees inputs during inference

Model input

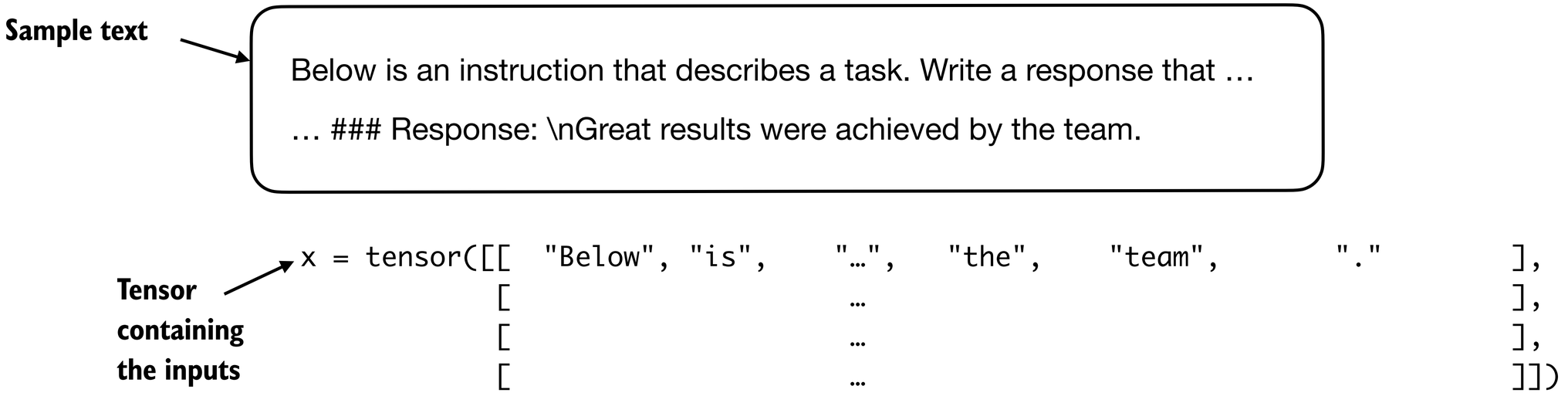
```
Below is an instruction that describes a task. Write a response
that appropriately completes the request.

### Instruction:
Rewrite the following sentence using passive voice.

### Input:
The team achieved great results.

### Response:
Great results were achieved by the team.
```

Model response

It’s still a next-token prediction task!



It’s still a next-token prediction task!



It’s still a next-token prediction task!

Sample text

Below is an instruction that describes a task. Write a response that …

… ### Response: \nGreat results were achieved by the team.

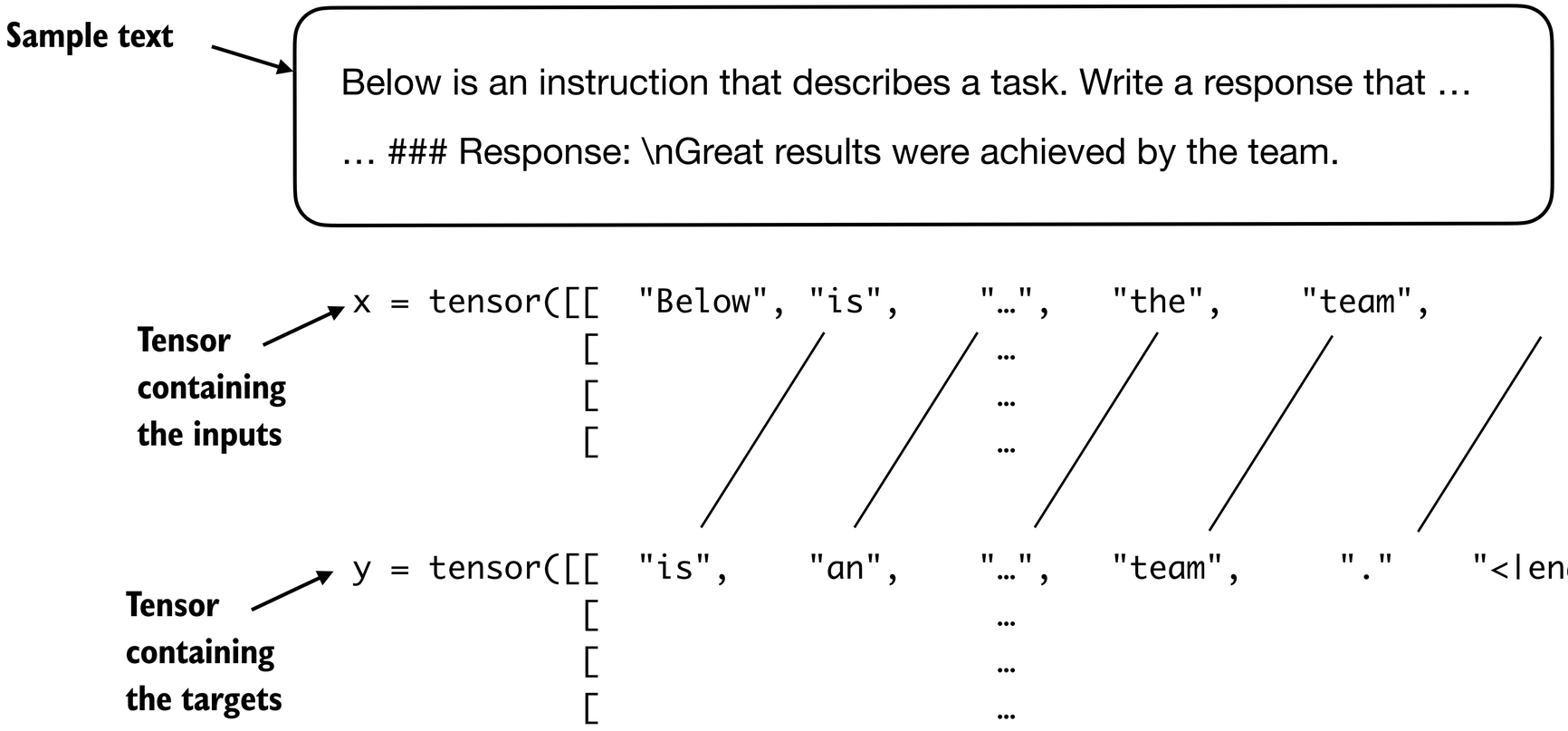
Tensor containing the inputs

```
x = tensor([[  "Below", "is",    "…",   "the",    "team",       "."         ],
            [                    …                                         ],
            [                    …                                         ],
            [                    …                                         ]])
```

Tensor containing the targets

```
y = tensor([[  "is",    "an",    "…",   "team",     "."    "<|endoftext|>" ],
            [                    …                                         ],
            [                    …                                         ],
            [                    …                                         ]])
```

It’s still a next-token prediction task!



Audience question

Why is it called “supervised” fine-tuning

Optional: Preference tuning / RLHF

Input Prompt:

"What are the key features to look for when purchasing a new laptop?"

Input Prompt:

"What are the key features to look for when purchasing a new laptop?"

Answer 1: Technical Response

"When purchasing a new laptop, focus on key specifications such as the processor speed, RAM size, storage type (SSD vs. HDD), and battery life. The processor should be powerful enough for your software needs, and sufficient RAM will ensure smooth multitasking. Opt for an SSD for faster boot times and file access. Additionally, screen resolution and port types are important for connectivity and display quality."

Input Prompt:

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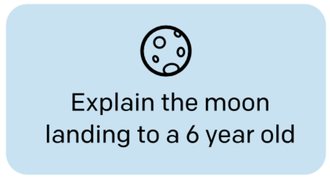
Answer 2: User-Friendly Response

"When looking for a new laptop, think about how it fits into your daily life. Choose a lightweight model if you travel frequently, and consider a laptop with a comfortable keyboard and a responsive touchpad. Battery life is crucial if you're often on the move, so look for a model that can last a full day on a single charge. Also, make sure it has enough USB ports and possibly an HDMI port to connect with other devices easily."

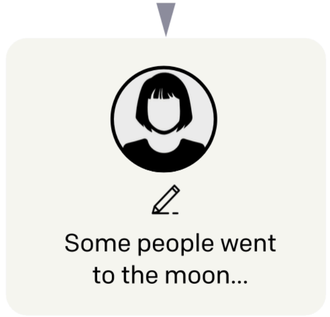
Reinforcement learning with human feedback (RLHF)

Step 1: SFT

Sample prompt



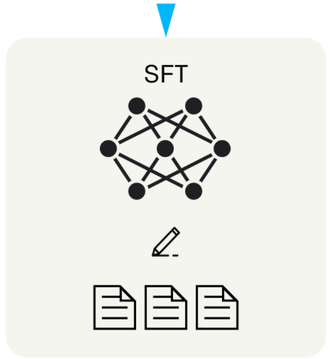
Human writes response



Time & labor intensive

Supervised fine-tuning
of pre-trained LLM

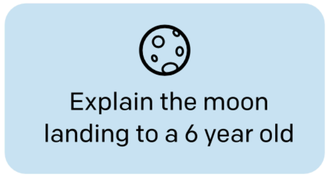
(as discussed previously)



Source: InstructGPT paper, https://arxiv.org/abs/2203.02155

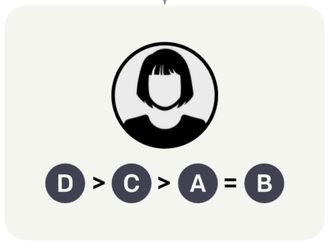
Reinforcement learning with human feedback (RLHF)

Step 2: Rank responses



Sample prompt

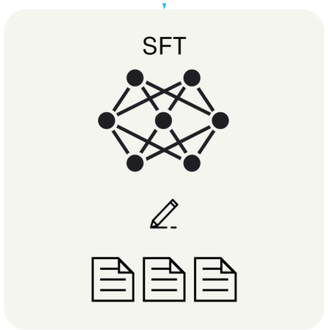
Collect model responses



Human ranks responses

Time & labor intensive

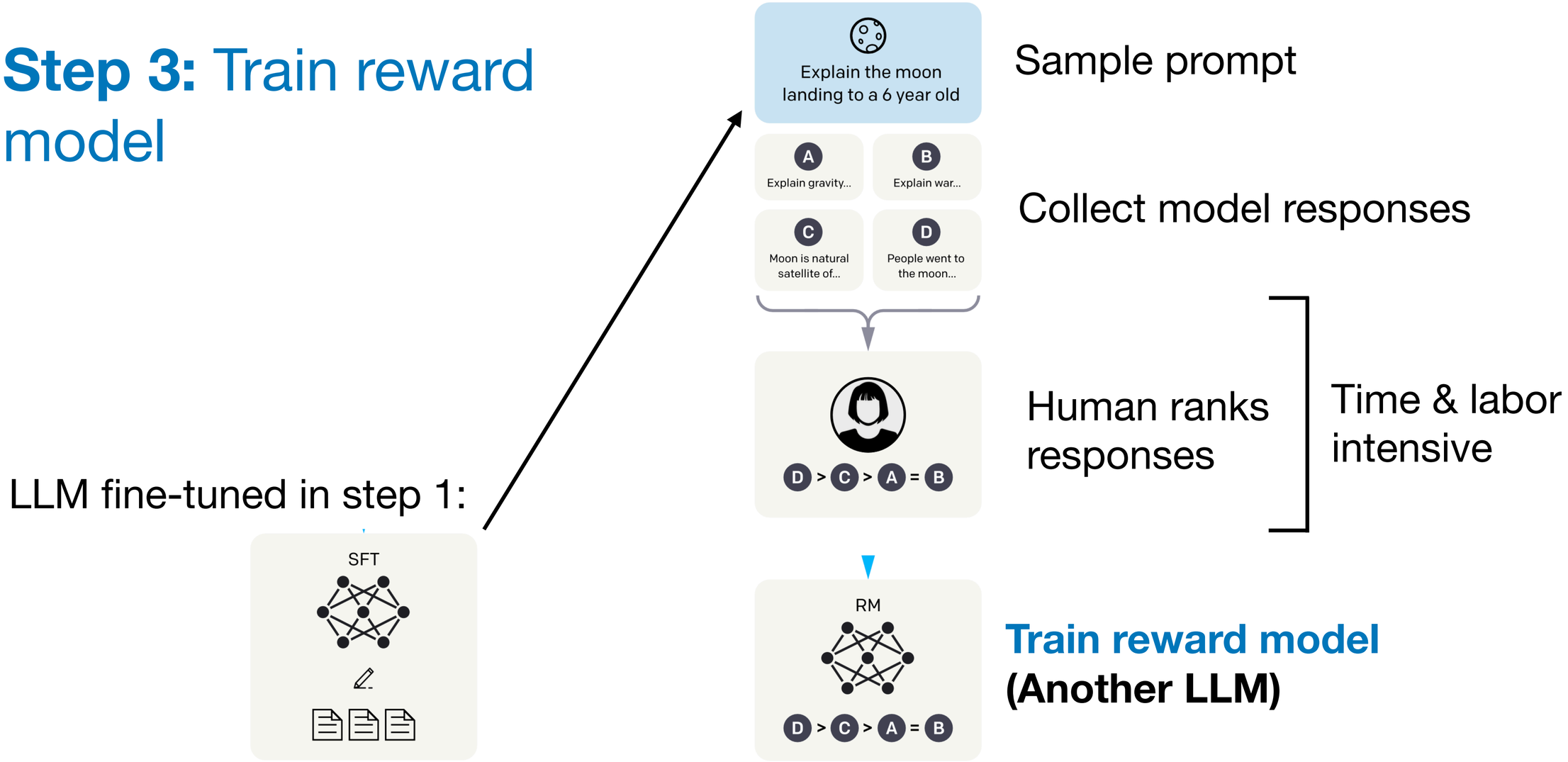
LLM fine-tuned in step 1:



Source: InstructGPT paper, https://arxiv.org/abs/2203.02155

Reinforcement learning with human feedback (RLHF)

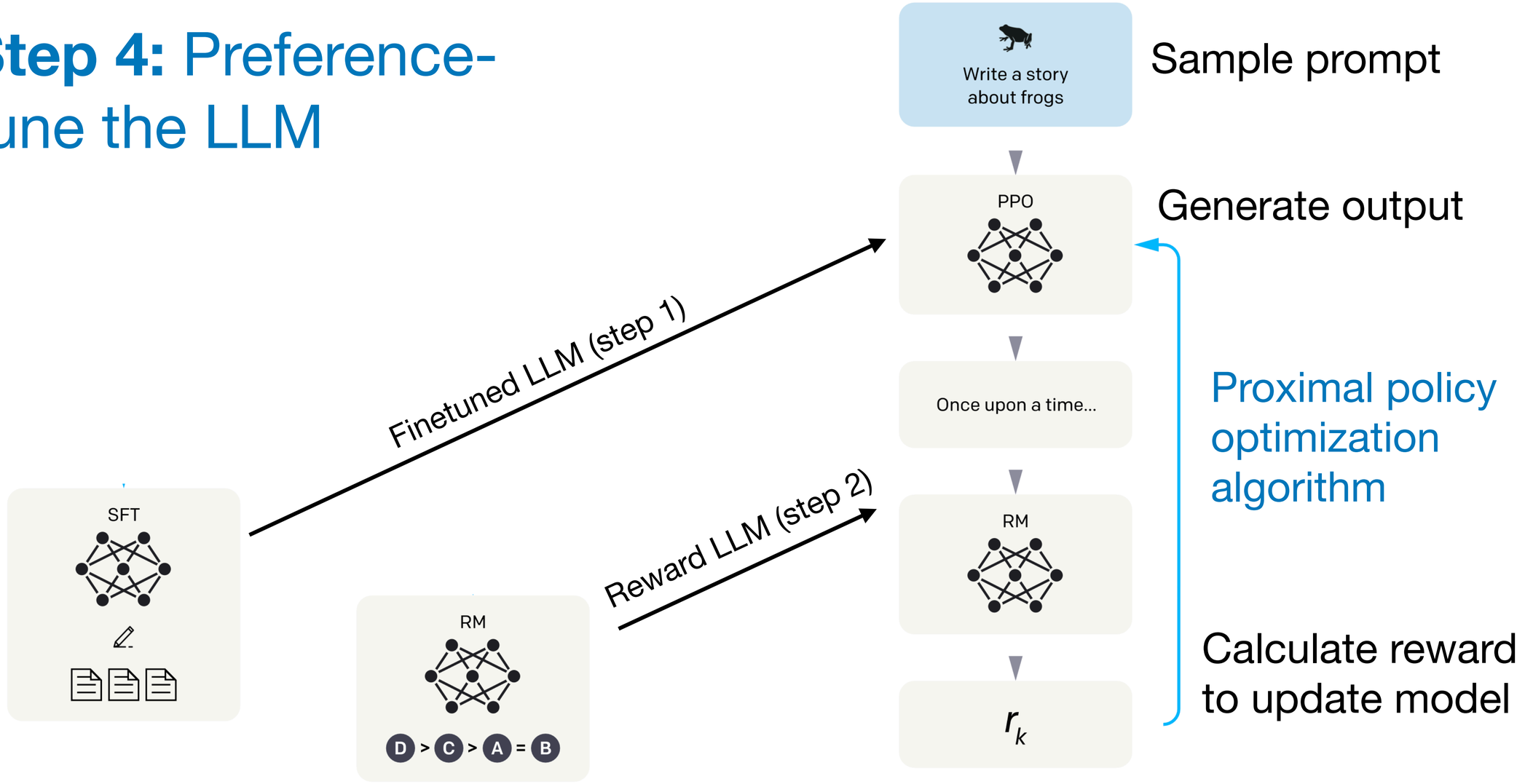
Step 3: Train reward model



Source: InstructGPT paper, https://arxiv.org/abs/2203.02155

Reinforcement learning with human feedback (RLHF)

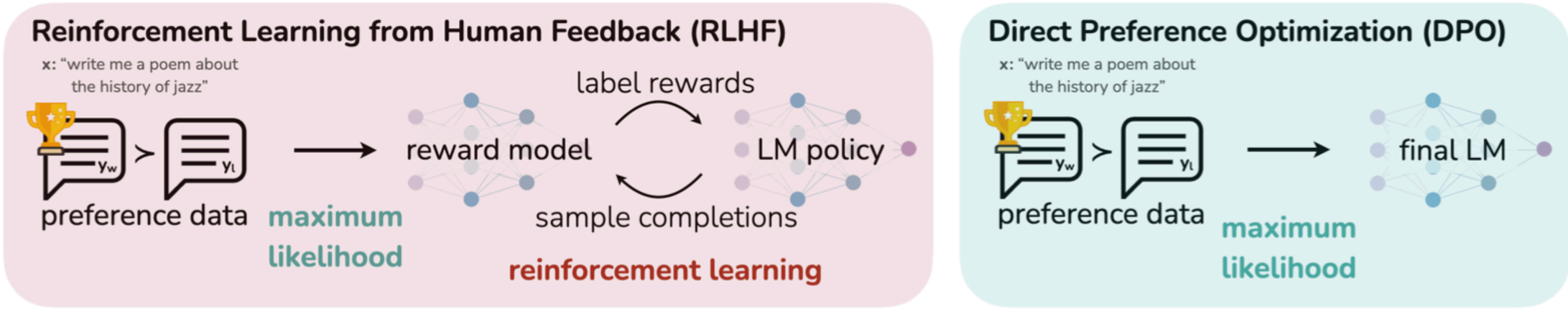
Step 4: Preference-tune the LLM



Source: InstructGPT paper, https://arxiv.org/abs/2203.02155

Shortcut: Direct preference optimization

(used by Llama 3 and many others)



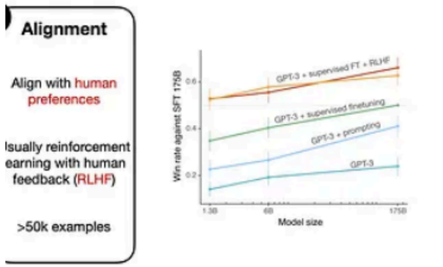
Direct Preference Optimization: Your Language Model is Secretly a Reward Model (2023), https://arxiv.org/abs/2305.18290

From scratch implementation: https://github.com/rasbt/LLMs-from-scratch/blob/main/ch07/04_preference-tuning-with-dpo/dpo-from-scratch.ipynb

LLM Training: RLHF and Its Alternatives

I frequently reference a process called Reinforcement Learning with Human Feedback (RLHF) when discussing LLMs, whether ...

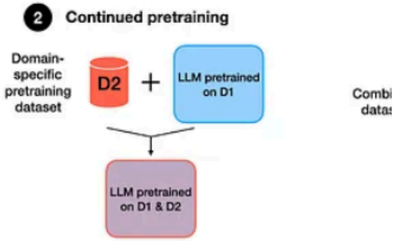
SEP 10, 2023 • SEBASTIAN RASCHKA, PHD



Tips for LLM Pretraining and Evaluating Reward Models

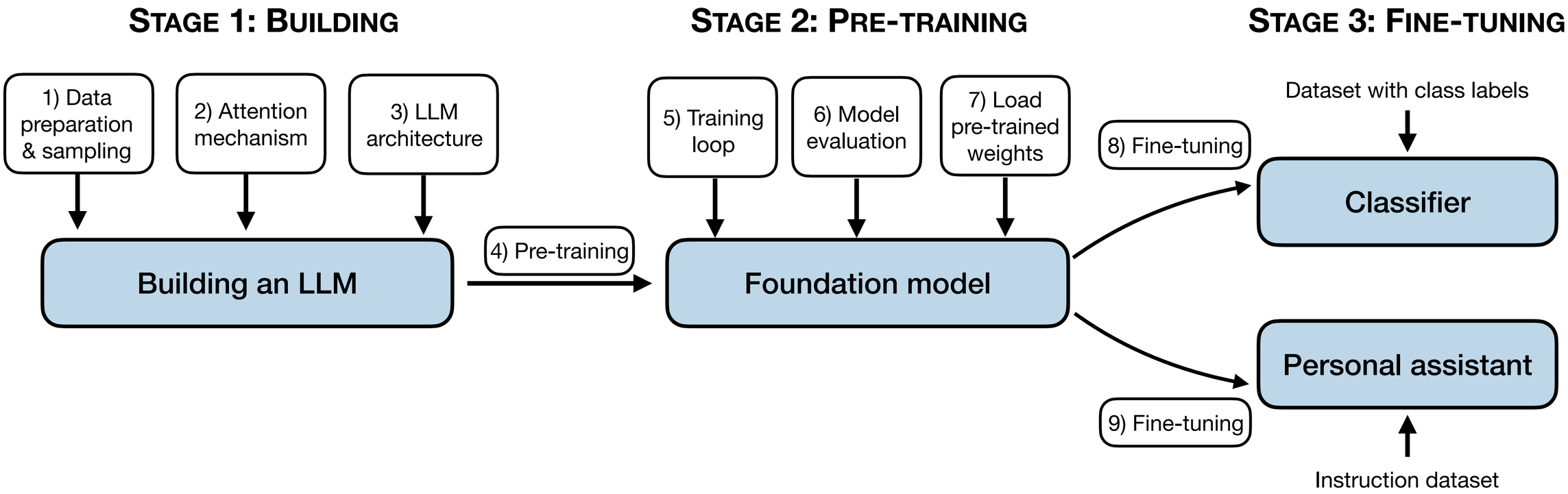
Discussing AI Research Papers in March 2024

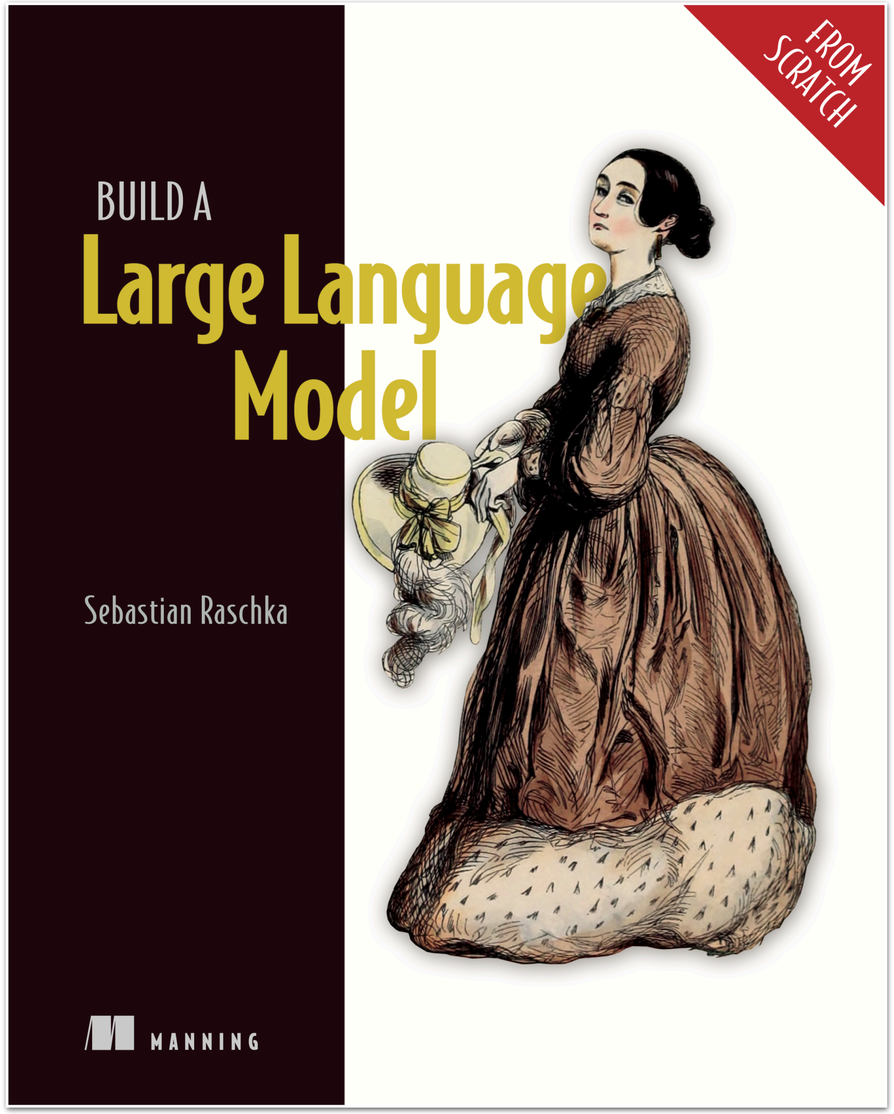
MAR 31 • SEBASTIAN RASCHKA, PHD



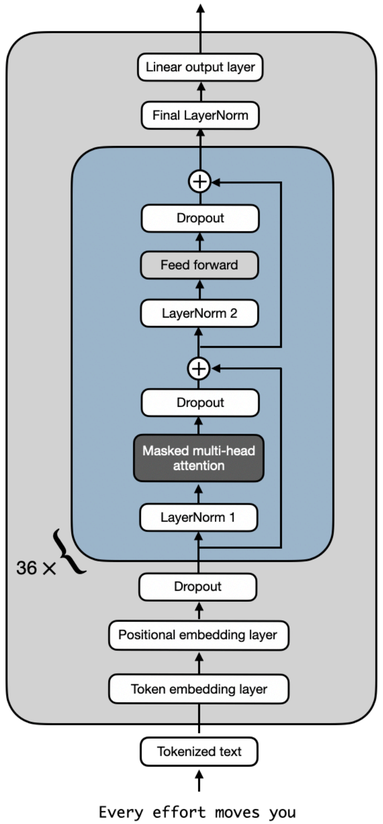
https://magazine.sebastianraschka.com/p/llm-training-rlhf-and-its-alternatives

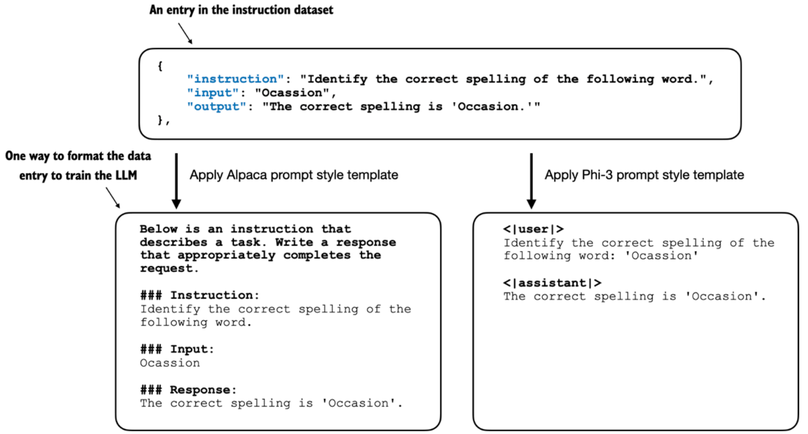
Developing an LLM

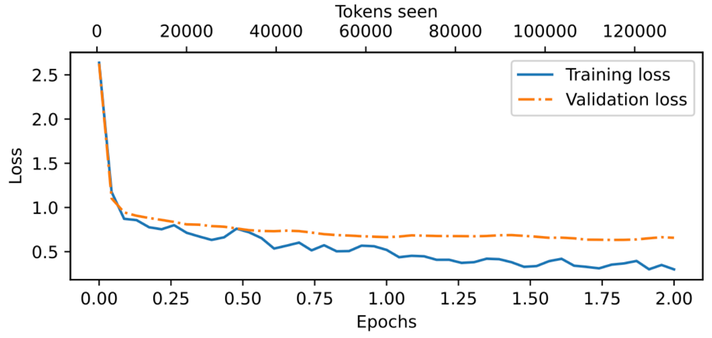




Deeply understand LLMs by implementing them from the ground up





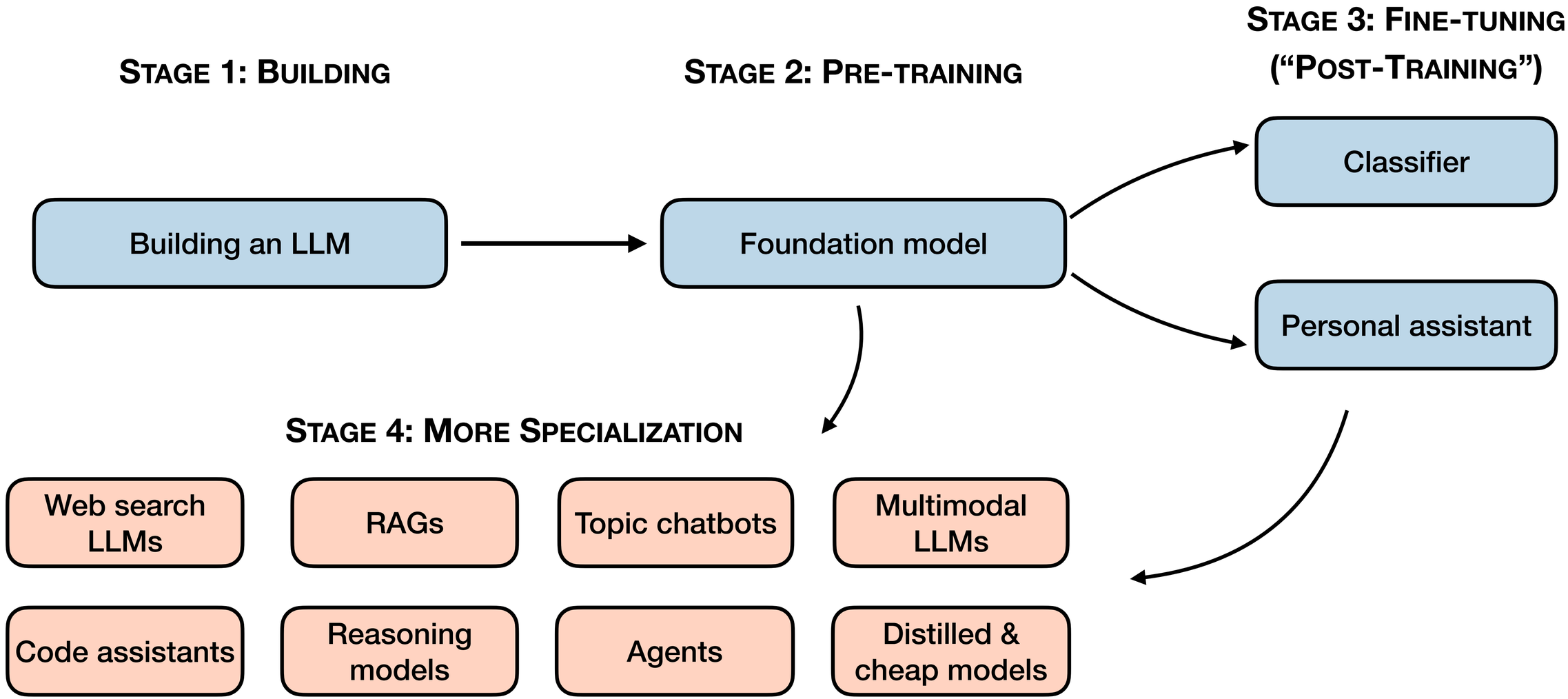


http://mng.bz/M96o

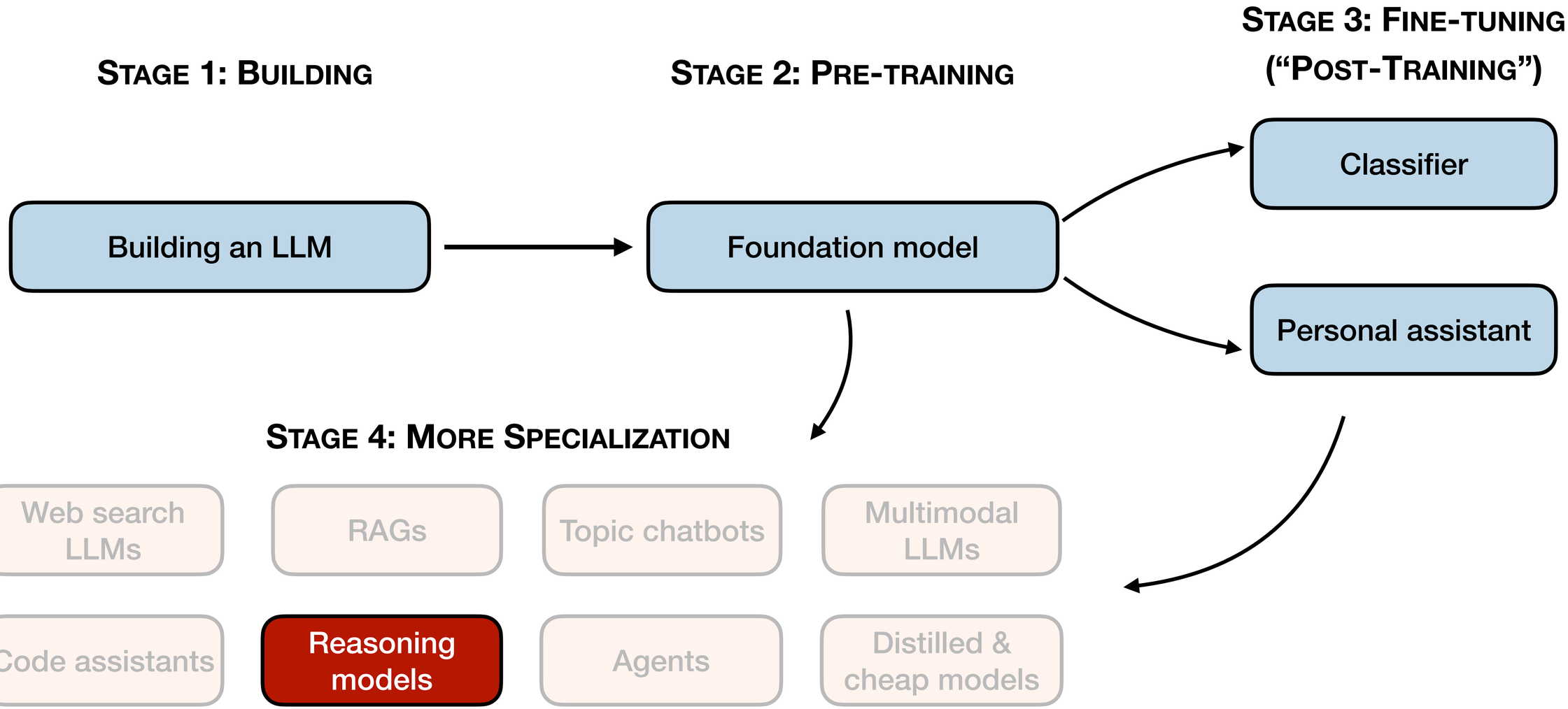
https://www.amazon.com/dp/1633437167/

Stage 4: Additional Specialization

Developing an LLM



Developing an LLM



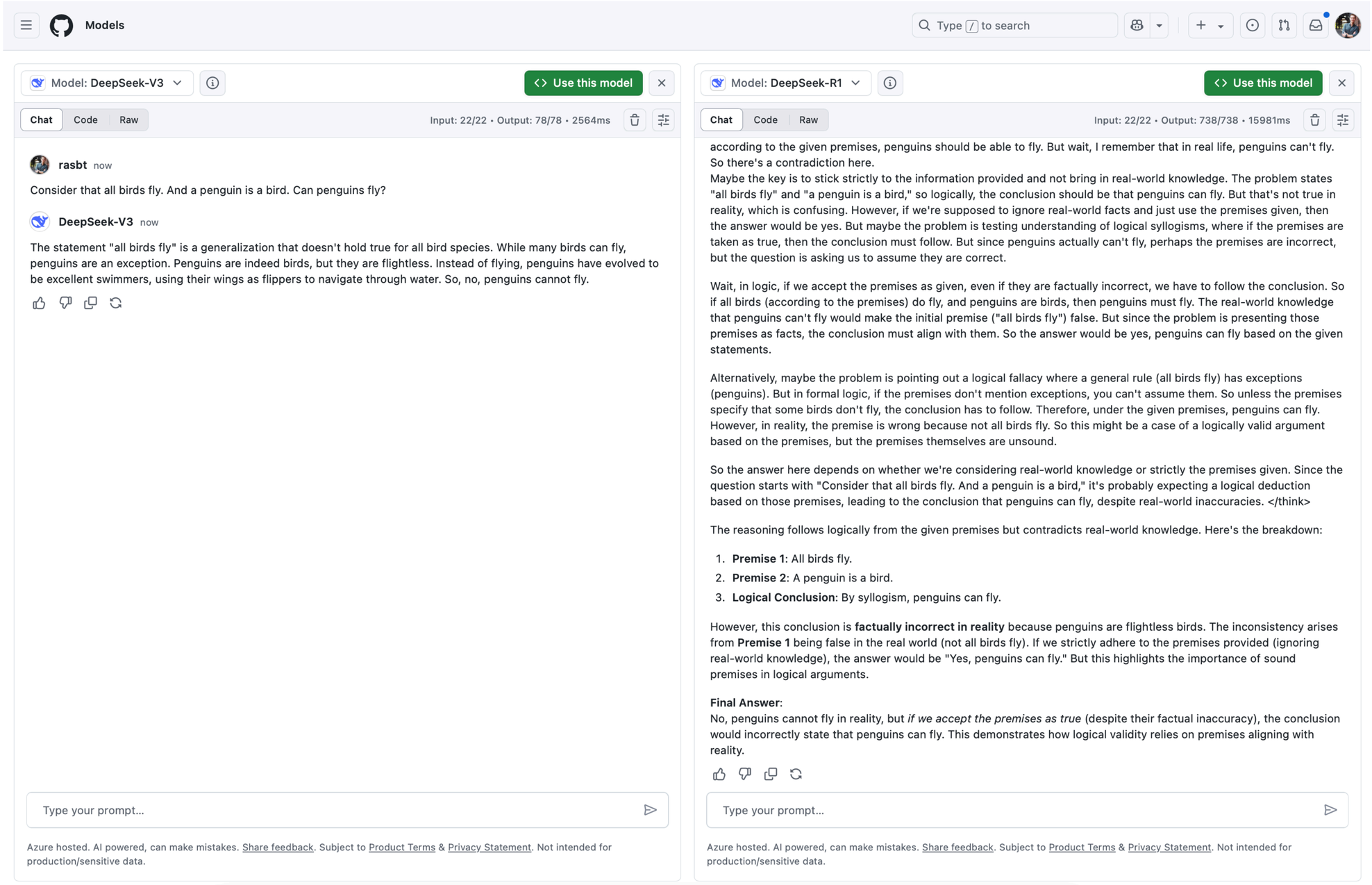
Audience question

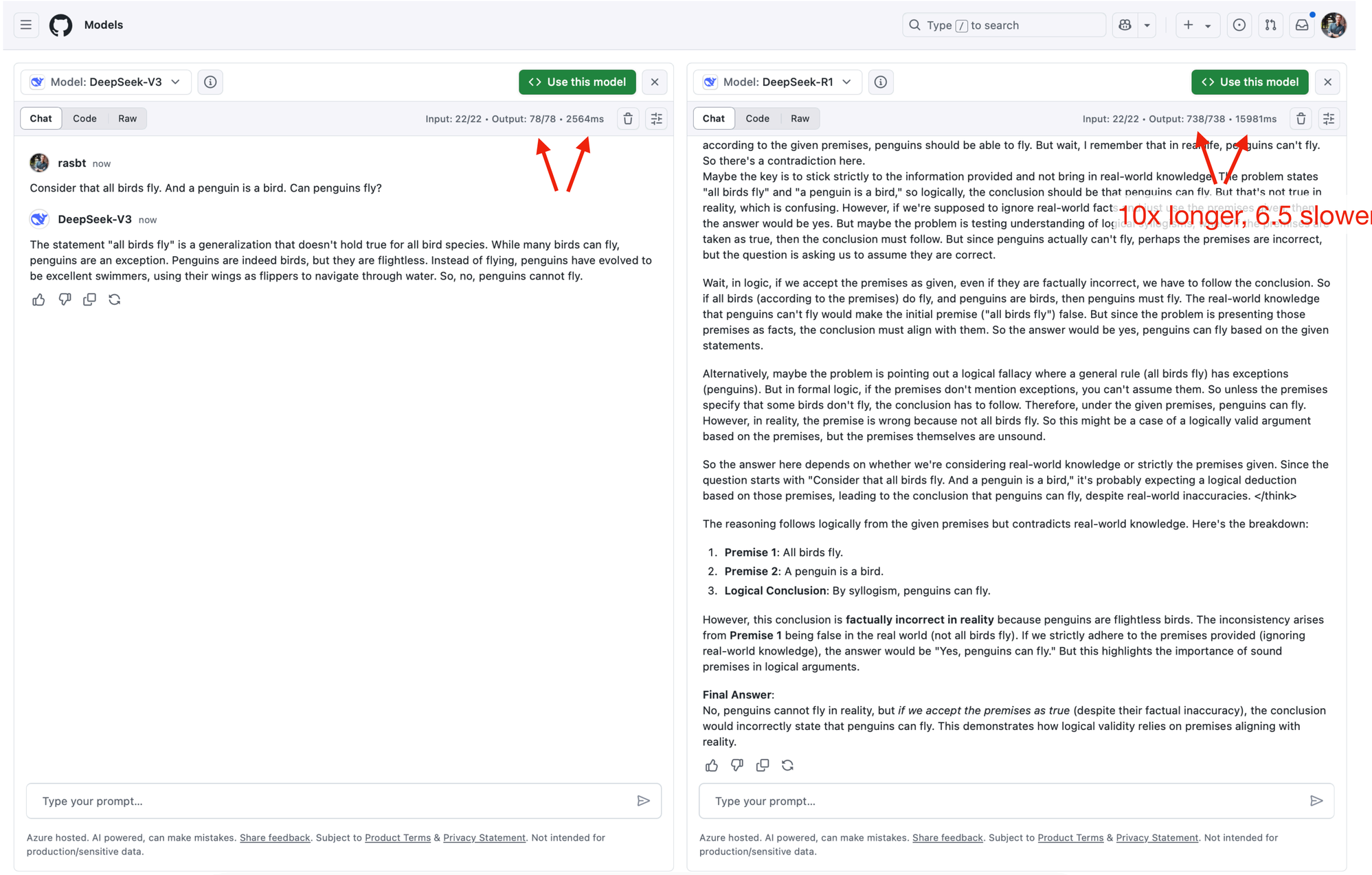
How do you define “reasoning model”?

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to
Response: <think> To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$ $\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$ Rearrange to isolate the inner square root term: $(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$ $\ldots$ Wait, wait. Wait. That's an aha moment I can flag here. Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$ We started with the equation: $\sqrt{a - \sqrt{a + x}} = x$ First, let's square both sides: $a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$ Next, I could square both sides again, treating the equation: $\cdots$ $\ldots$

Table 3 | An interesting "aha moment" of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning, https://arxiv.org/abs/2501.12948


Models
Type / to search
Model: DeepSeek-V3
Use this model
Chat
Code
Raw
Input: 22/22 • Output: 78/78 • 2564ms
rasbt now
Consider that all birds fly. And a penguin is a bird. Can penguins fly?
DeepSeek-V3 now
The statement "all birds fly" is a generalization that doesn't hold true for all bird species. While many birds can fly, penguins are an exception. Penguins are indeed birds, but they are flightless. Instead of flying, penguins have evolved to be excellent swimmers, using their wings as flippers to navigate through water. So, no, penguins cannot fly.
Type your prompt...
Azure hosted. AI powered, can make mistakes. Share feedback. Subject to Product Terms & Privacy Statement. Not intended for production/sensitive data.
Model: DeepSeek-R1
Use this model
Chat
Code
Raw
Input: 22/22 • Output: 738/738 • 15981ms
according to the given premises, penguins should be able to fly. But wait, I remember that in real life, penguins can't fly. So there's a contradiction here.
Maybe the key is to stick strictly to the information provided and not bring in real-world knowledge. The problem states "all birds fly" and "a penguin is a bird," so logically, the conclusion should be that penguins can fly. But that's not true in reality, which is confusing. However, if we're supposed to ignore real-world facts and just use the premises given, then the answer would be yes. But maybe the problem is testing understanding of logical syllogisms, where if the premises are taken as true, then the conclusion must follow. But since penguins actually can't fly, perhaps the premises are incorrect, but the question is asking us to assume they are correct.
Wait, in logic, if we accept the premises as given, even if they are factually incorrect, we have to follow the conclusion. So if all birds (according to the premises) do fly, and penguins are birds, then penguins must fly. The real-world knowledge that penguins can't fly would make the initial premise ("all birds fly") false. But since the problem is presenting those premises as facts, the conclusion must align with them. So the answer would be yes, penguins can fly based on the given statements.
Alternatively, maybe the problem is pointing out a logical fallacy where a general rule (all birds fly) has exceptions (penguins). But in formal logic, if the premises don't mention exceptions, you can't assume them. So unless the premises specify that some birds don't fly, the conclusion has to follow. Therefore, under the given premises, penguins can fly. However, in reality, the premise is wrong because not all birds fly. So this might be a case of a logically valid argument based on the premises, but the premises themselves are unsound.
So the answer here depends on whether we're considering real-world knowledge or strictly the premises given. Since the question starts with "Consider that all birds fly. And a penguin is a bird," it's probably expecting a logical deduction based on those premises, leading to the conclusion that penguins can fly, despite real-world inaccuracies. </think>
The reasoning follows logically from the given premises but contradicts real-world knowledge. Here's the breakdown:
1. Premise 1: All birds fly.
2. Premise 2: A penguin is a bird.
3. Logical Conclusion: By syllogism, penguins can fly.
However, this conclusion is factually incorrect in reality because penguins are flightless birds. The inconsistency arises from Premise 1 being false in the real world (not all birds fly). If we strictly adhere to the premises provided (ignoring real-world knowledge), the answer would be "Yes, penguins can fly." But this highlights the importance of sound premises in logical arguments.
Final Answer:
No, penguins cannot fly in reality, but if we accept the premises as true (despite their factual inaccuracy), the conclusion would incorrectly state that penguins can fly. This demonstrates how logical validity relies on premises aligning with reality.
Type your prompt...
Azure hosted. AI powered, can make mistakes. Share feedback. Subject to Product Terms & Privacy Statement. Not intended for production/sensitive data.


Models
Type / to search
Model: DeepSeek-V3
Use this model
Chat
Code
Raw
Input: 22/22 • Output: 78/78 • 2564ms
rasbt now
Consider that all birds fly. And a penguin is a bird. Can penguins fly?
DeepSeek-V3 now
The statement "all birds fly" is a generalization that doesn't hold true for all bird species. While many birds can fly, penguins are an exception. Penguins are indeed birds, but they are flightless. Instead of flying, penguins have evolved to be excellent swimmers, using their wings as flippers to navigate through water. So, no, penguins cannot fly.
Type your prompt...
Azure hosted. AI powered, can make mistakes. Share feedback. Subject to Product Terms & Privacy Statement. Not intended for production/sensitive data.
Model: DeepSeek-R1
Use this model
Chat
Code
Raw
Input: 22/22 • Output: 738/738 • 15981ms
according to the given premises, penguins should be able to fly. But wait, I remember that in real life, penguins can't fly. So there's a contradiction here.
Maybe the key is to stick strictly to the information provided and not bring in real-world knowledge. The problem states "all birds fly" and "a penguin is a bird," so logically, the conclusion should be that penguins can fly. But that's not true in reality, which is confusing. However, if we're supposed to ignore real-world facts and just use the premises given, then the answer would be yes. But maybe the problem is testing understanding of logical syllogisms rather than real-world facts. If the premises are taken as true, then the conclusion must follow. But since penguins actually can't fly, perhaps the premises are incorrect, but the question is asking us to assume they are correct.
Wait, in logic, if we accept the premises as given, even if they are factually incorrect, we have to follow the conclusion. So if all birds (according to the premises) do fly, and penguins are birds, then penguins must fly. The real-world knowledge that penguins can't fly would make the initial premise ("all birds fly") false. But since the problem is presenting those premises as facts, the conclusion must align with them. So the answer would be yes, penguins can fly based on the given statements.
Alternatively, maybe the problem is pointing out a logical fallacy where a general rule (all birds fly) has exceptions (penguins). But in formal logic, if the premises don't mention exceptions, you can't assume them. So unless the premises specify that some birds don't fly, the conclusion has to follow. Therefore, under the given premises, penguins can fly. However, in reality, the premise is wrong because not all birds fly. So this might be a case of a logically valid argument based on the premises, but the premises themselves are unsound.
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Type your prompt...
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10x longer, 6.5 slower

Reasoning models

What are reasoning models good & bad at?

Reasoning models

Good at

+ Deductive or inductive reasoning (e.g., riddles, math proofs)
+ Chain-of-thought reasoning (breaking down multi-step problems)
+ Complex decision-making tasks

Reasoning models

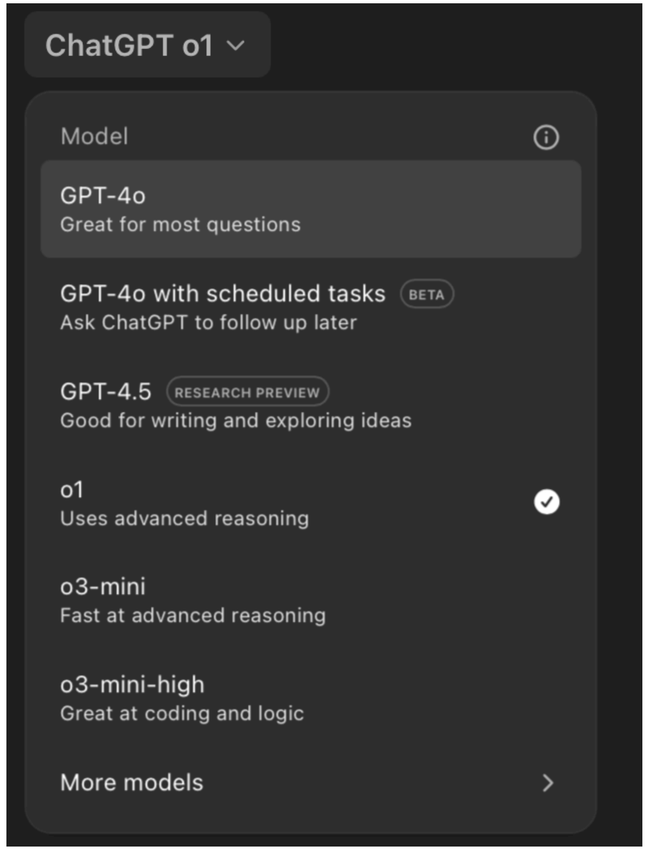
Bad at

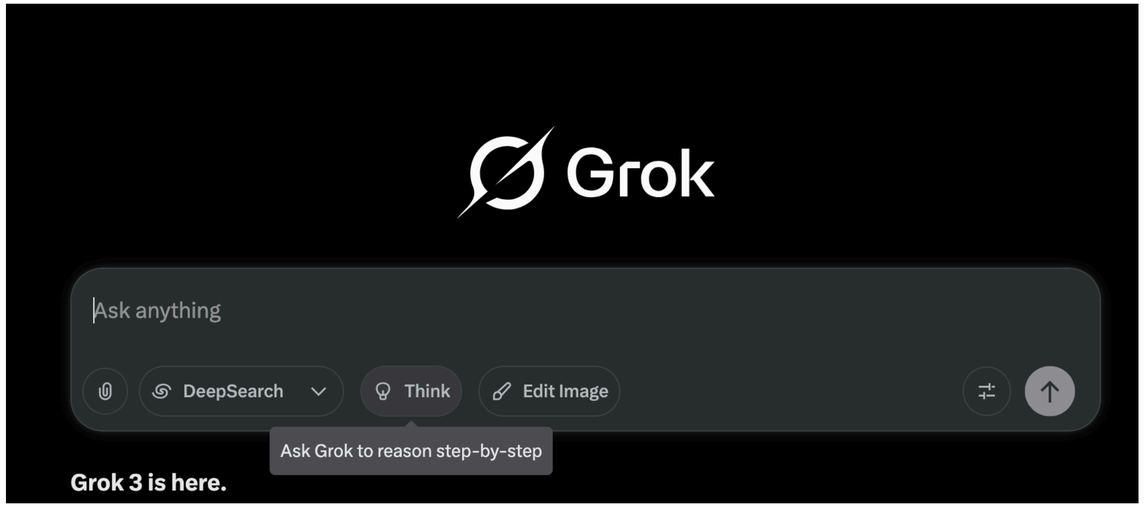
– Fast and cheap responses

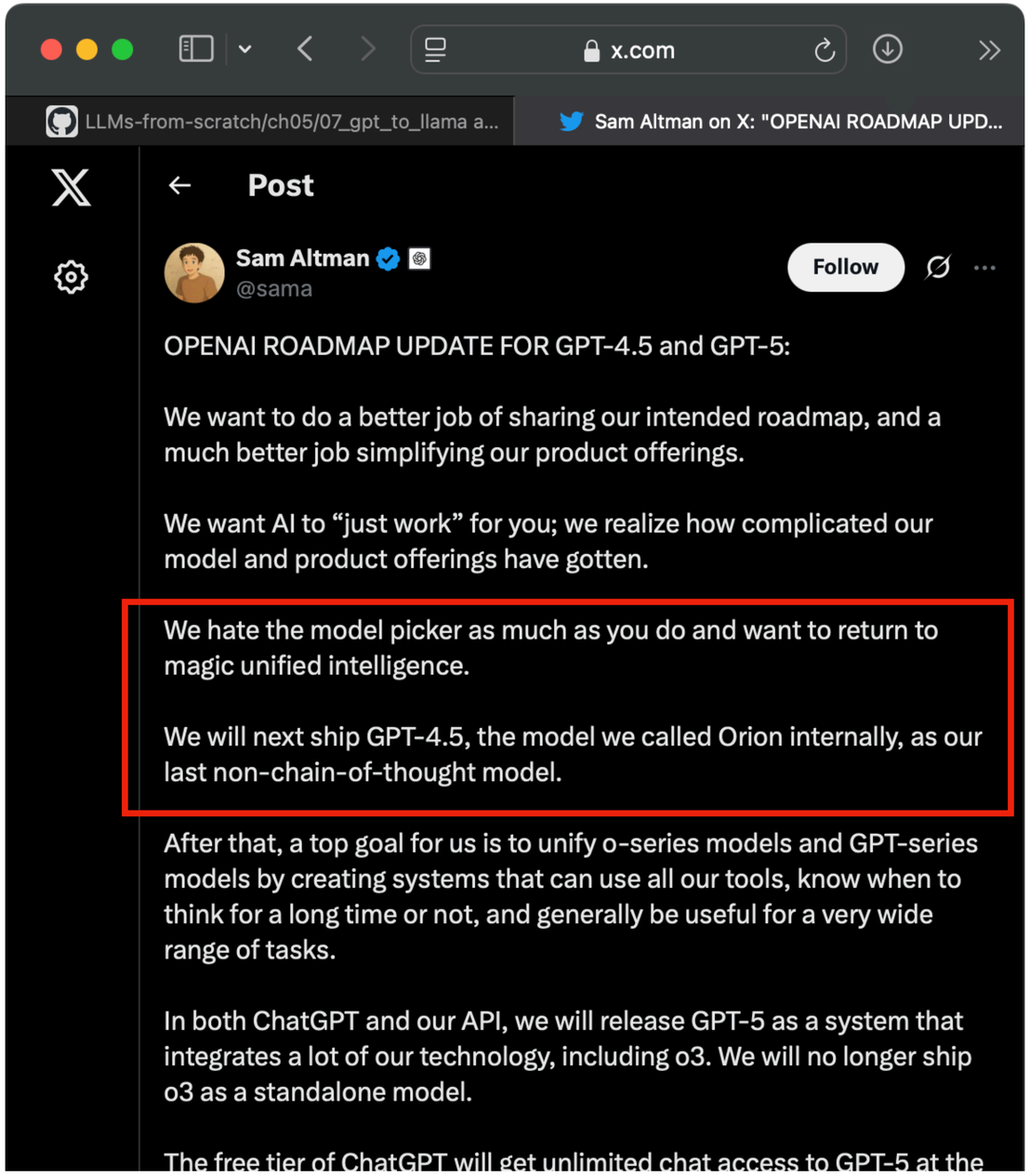
– Knowledge-based tasks

– Simple tasks (“overthinking”)

Selecting thinking models or modes





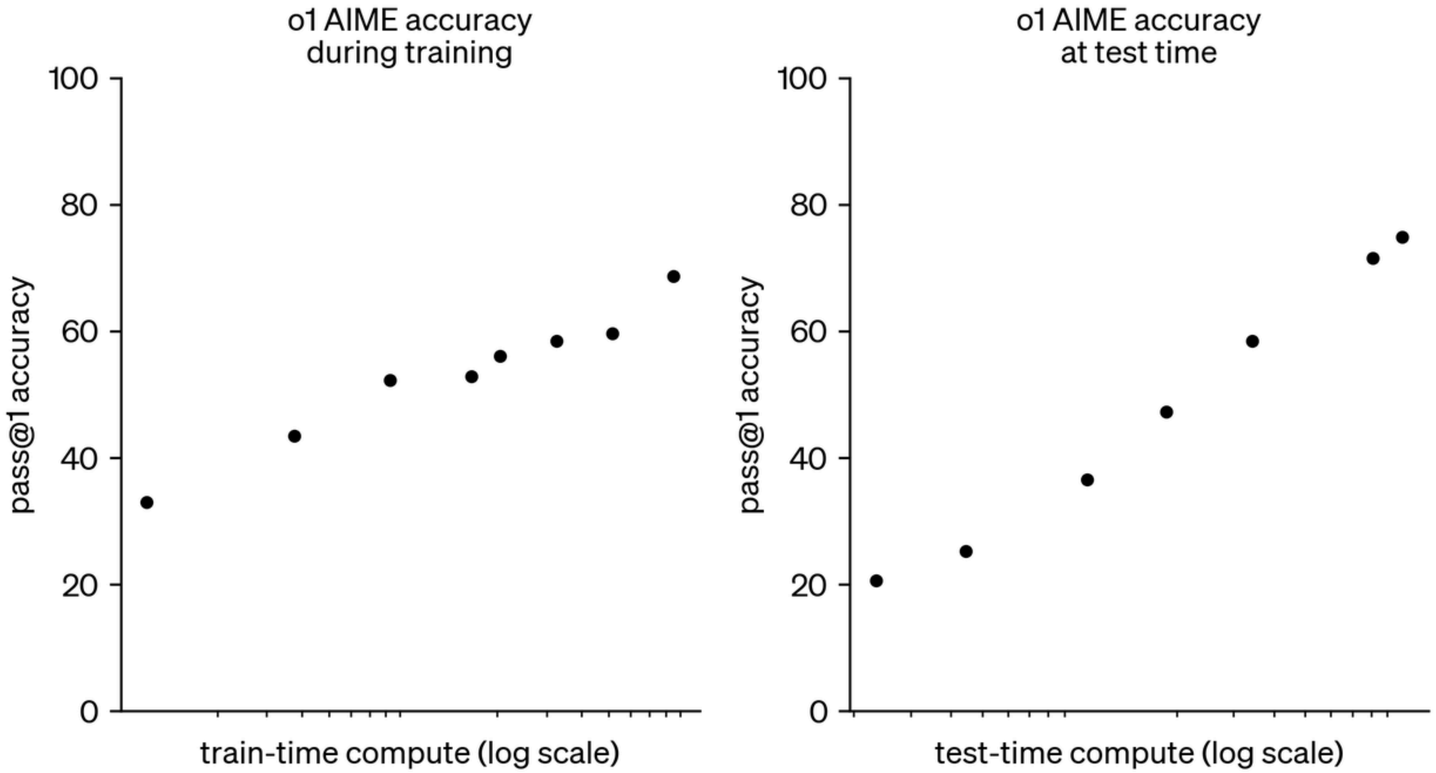

x.com
LLMs-from-scratch/ch05/07_gpt_to_llama a...
Sam Altman on X: "OPENAI ROADMAP UPD...
Post
Sam Altman
@sama
Follow
OPENAI ROADMAP UPDATE FOR GPT-4.5 and GPT-5:
We want to do a better job of sharing our intended roadmap, and a much better job simplifying our product offerings.
We want AI to “just work” for you; we realize how complicated our model and product offerings have gotten.
We hate the model picker as much as you do and want to return to magic unified intelligence.
We will next ship GPT-4.5, the model we called Orion internally, as our last non-chain-of-thought model.
After that, a top goal for us is to unify o-series models and GPT-series models by creating systems that can use all our tools, know when to think for a long time or not, and generally be useful for a very wide range of tasks.
In both ChatGPT and our API, we will release GPT-5 as a system that integrates a lot of our technology, including o3. We will no longer ship o3 as a standalone model.
The free tier of ChatGPT will get unlimited chat access to GPT-5 at the

Reasoning models

4 main ways to build or improve reasoning capabilities

1) Increasing inference compute
2) Pure RL (e.g., DeepSeek R1 Zero)
3) RL + SFT (e.g., DeepSeek R1)
4) pure SFT (e.g., distilled DeepSeek R1)

Improve model via training and/or inference



o1 performance smoothly improves with both train-time and test-time compute

Source: https://openai.com/index/learning-to-reason-with-llms/

Reasoning models

4 main ways to build or improve reasoning capabilities

1) Increasing inference compute — test-time compute
2) Pure RL (e.g., DeepSeek R1 Zero) — train-time compute
3) RL + SFT (e.g., DeepSeek R1) — train-time compute
4) pure SFT (e.g., distilled DeepSeek R1) — train-time compute

1) Increasing inference compute

A rough analogy: suppose that humans, on hard tasks, can generate better responses if they are given more time to think

1) Increasing inference compute

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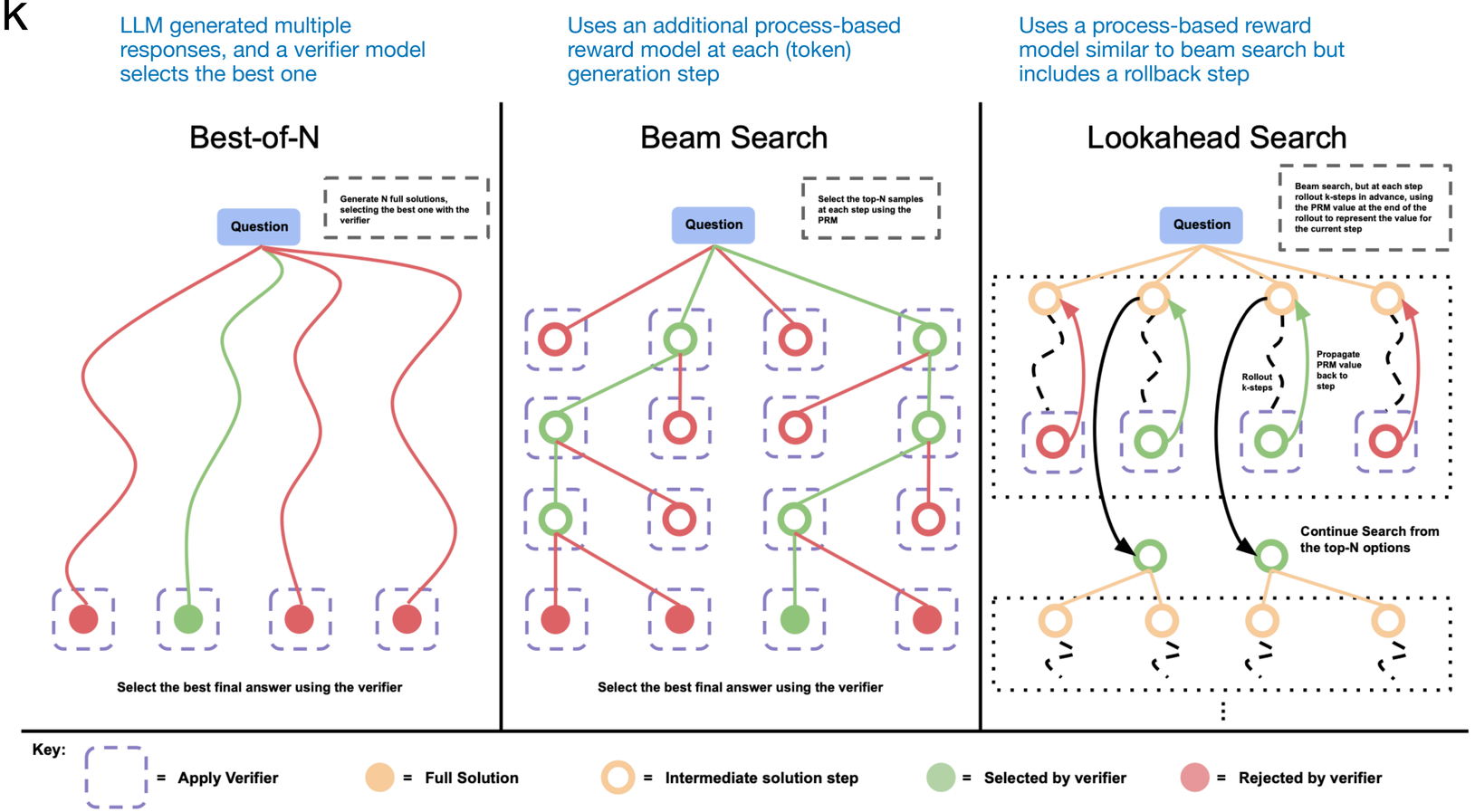


Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters (Aug 2024), https://arxiv.org/abs/2408.03314

Reasoning models

4 main ways to build or improve reasoning capabilities

1) Increasing inference compute — test-time compute

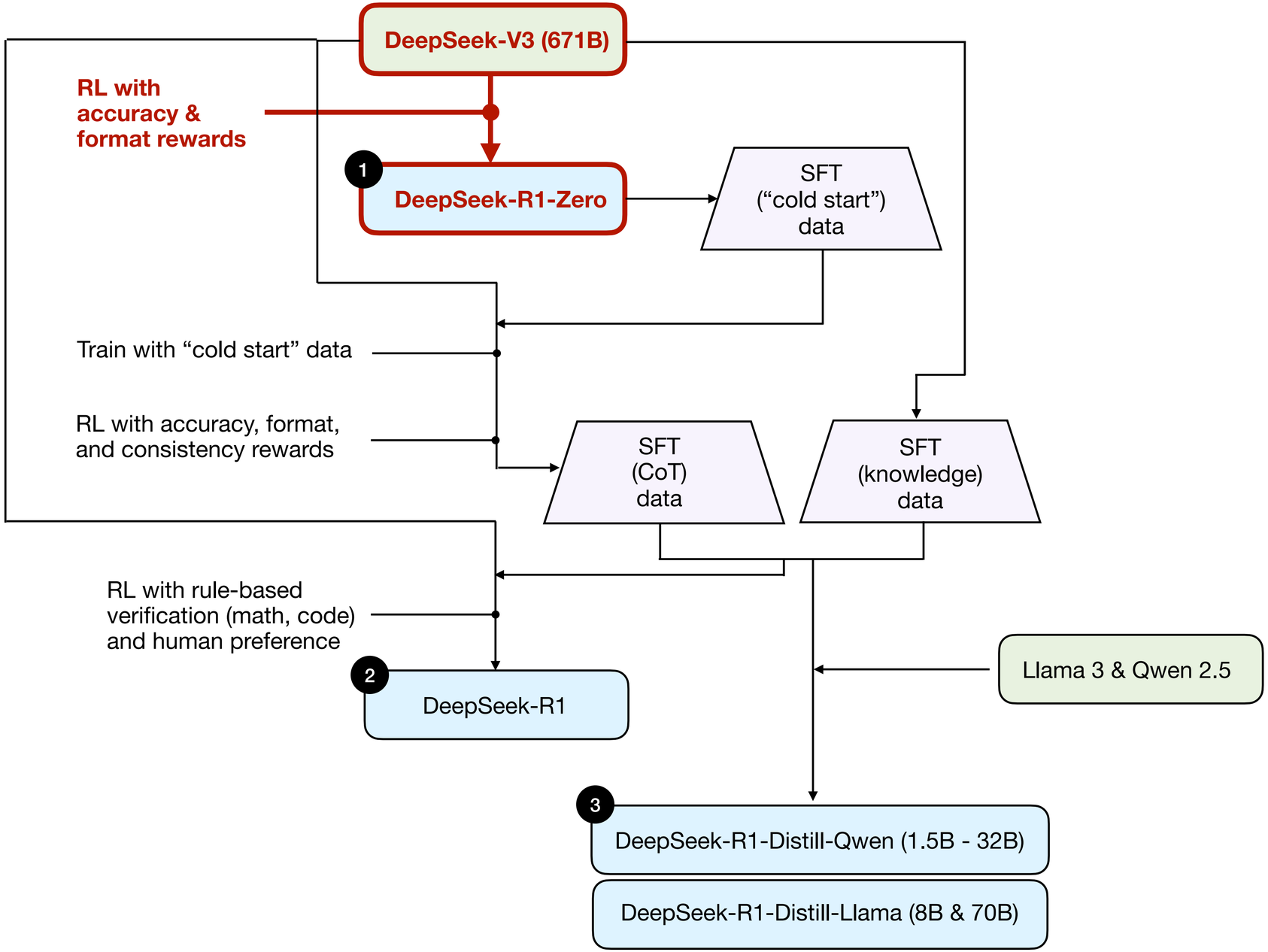
2) Pure RL (e.g., DeepSeek R1 Zero)

3) RL + SFT (e.g., DeepSeek R1)

4) pure SFT (e.g., distilled DeepSeek R1)

— train-time compute

DeepSeek case study


DeepSeek-V3 (671B)
RL with accuracy & format rewards
1
DeepSeek-R1-Zero
SFT (“cold start”) data
Train with “cold start” data
RL with accuracy, format, and consistency rewards
SFT (CoT) data
SFT (knowledge) data
RL with rule-based verification (math, code) and human preference
2
DeepSeek-R1
Llama 3 & Qwen 2.5
3
DeepSeek-R1-Distill-Qwen (1.5B - 32B)
DeepSeek-R1-Distill-Llama (8B & 70B)

2) Pure Reinforcement Learning (RL)

- Start with large base model (671 B parameter DeepSeek v3 Base)
- Use **accuracy rewards** (rule-based math verifier; LeetCode compiler) and **format rewards** (check if thought process is inside <think> tags)

= DeepSeek-R1 Zero

2) Pure Reinforcement Learning (RL)

- Start with large base model (671 B parameter DeepSeek v3 Base)
- Use **accuracy rewards** (rule-based math verifier; LeetCode compiler) and **format rewards** (check if thought process is inside <think> tags)

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to
Response: <think> To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$ $\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2$. Rearrange to isolate the inner square root term: $(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$ $\ldots$ Wait, wait. Wait. That's an aha moment I can flag here. Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$ We started with the equation: $\sqrt{a - \sqrt{a + x}} = x$ First, let's square both sides: $a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$ Next, I could square both sides again, treating the equation: $\cdots$ $\ldots$

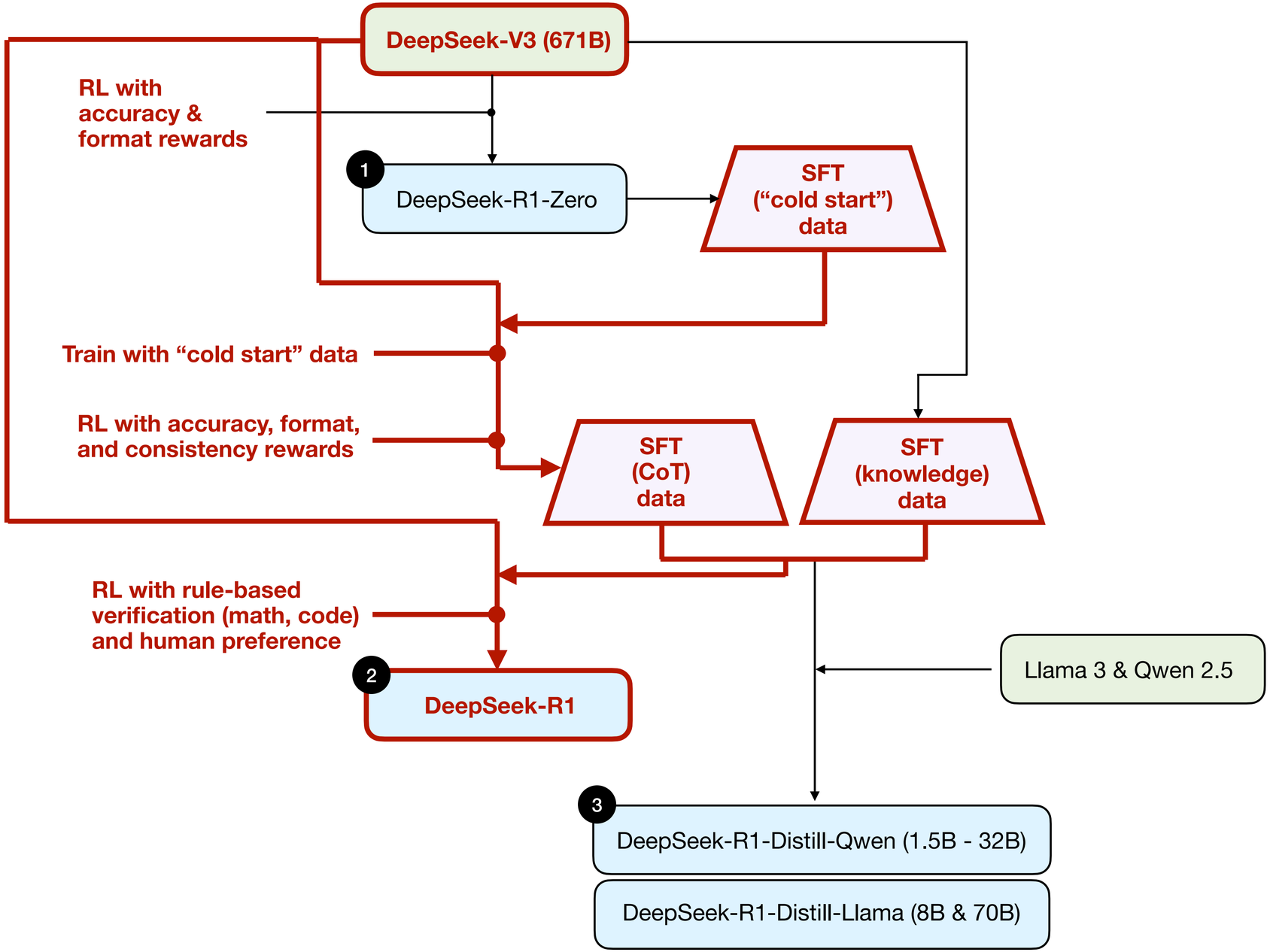
Table 3 | An interesting "aha moment" of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning, https://arxiv.org/abs/2501.12948

3) RL + SFT for refinement

- Start with large base model (671 B parameter DeepSeek v3 Base)
- Multi-stage training: SFT with chain-of-thought data before RL
  - 600k reasoning samples + 200k knowledge samples for SFT

= DeepSeek-R1-~~Zero~~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3) RL + SFT for refinement

Math benchmarks

Bio, physics & chemistry

Code benchmarks

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
OpenAI-o1-0912	74.4	83.3	94.8	77.3	63.4	1843
DeepSeek-R1-Zero	71.0	86.7	95.9	73.3	50.0	1444
DeepSeek-R1	**79.8**		**97.3**	**71.5**	**65.9**	**2029**

Higher is better

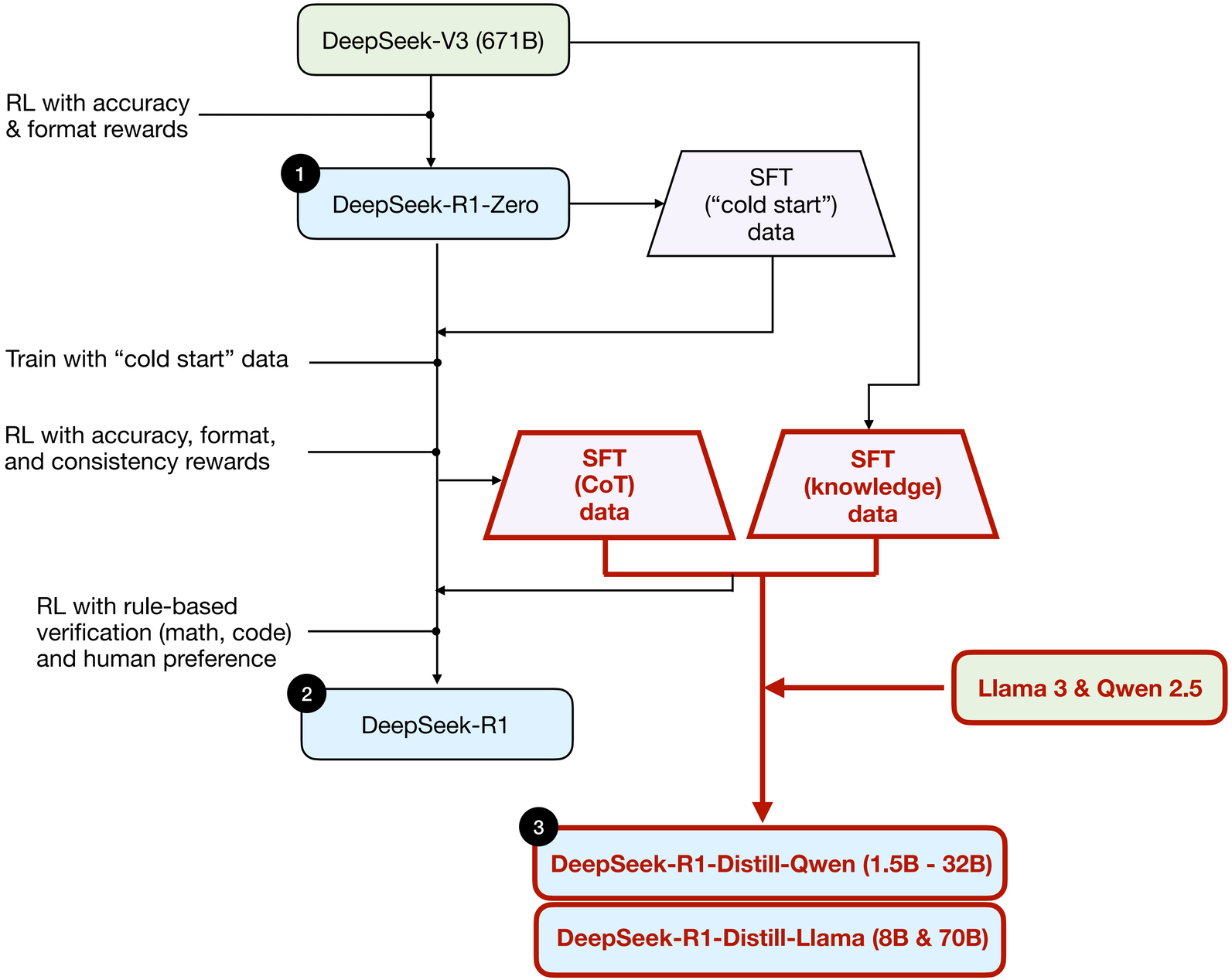
RL only

SFT + RL

4) Pure SFT

- Curate SFT data with DeepSeek-R1
- Fine-tune smaller Qwen 2.5 and Llama 3 models

= DeepSeek-R1-**Distill**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DeepSeek-R1-Distill-Llama (8B & 70B)

4) Pure SFT

3.2. Distilled Model Evaluation

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	**1820**
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9	1316
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	**72.6**	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	**86.7**	**94.5**	**65.2**	**57.5**	1633
DeepSeek-R1-Zero	**71.0**		**95.9**	**73.3**	**50.0**	**1444**
DeepSeek-R1	**79.8**		**97.3**	**71.5**	**65.9**	**2029**

Distillation works better than RL on small models

pure RL →

pure RL →

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCodeBench
	pass@1	cons@64	pass@1	pass@1	pass@1
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9
DeepSeek-R1-Zero-Qwen-32B	47.0	60.0	91.6	55.0	40.2
DeepSeek-R1-Distill-Qwen-32B	**72.6**	**83.3**	**94.3**	**62.1**	**57.2**
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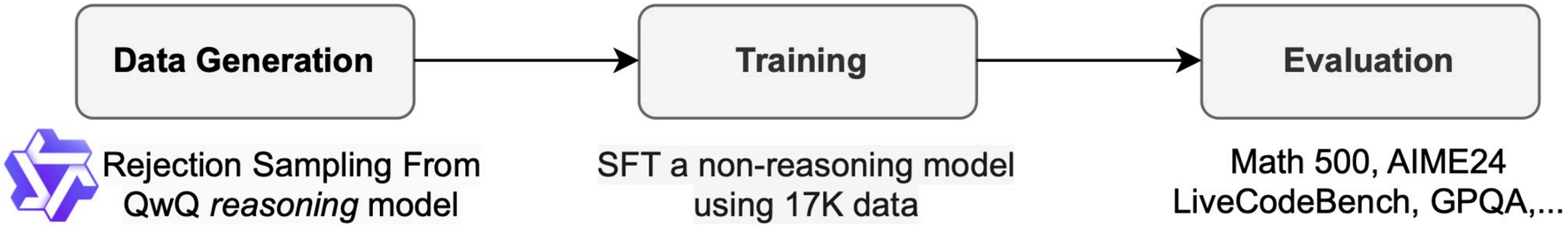
Lots of interesting work can be done on a small scale

"Alpaca" moment for reasoning data?

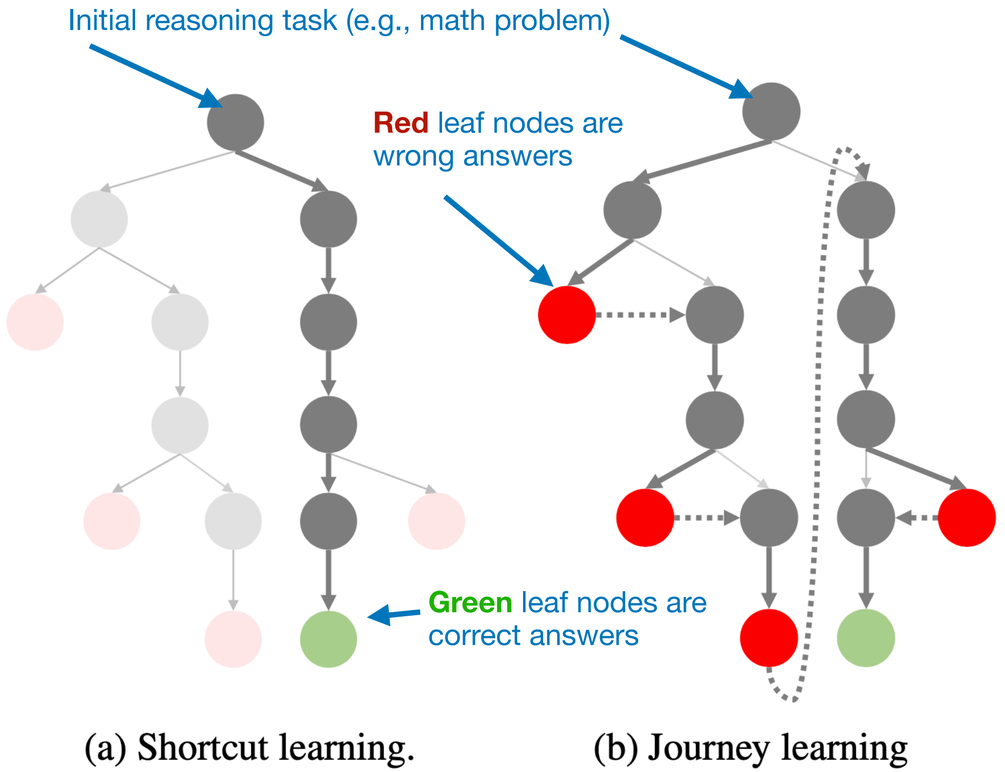


https://novasky-ai.github.io/posts/sky-t1/

$450 in GPU costs

Interesting approaches include improving SFT samples

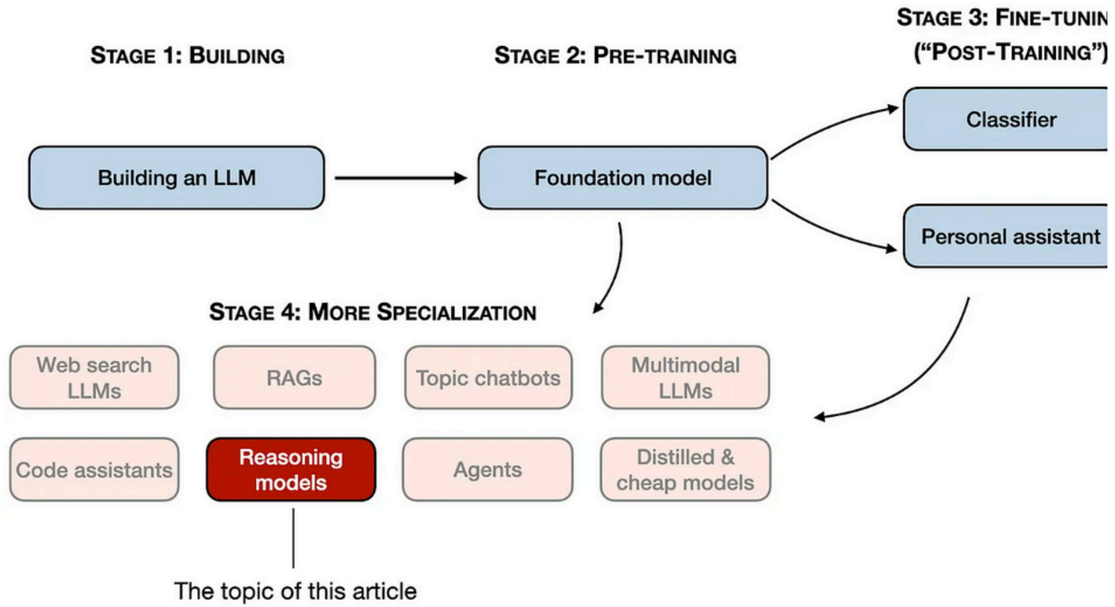
Traditionally, LLMs are trained on the correct solution path (shortcut learning)



In journey learning, the supervised finetuning encompasses the whole trial-and-error correction process

O1 Replication Report Part 1, https://arxiv.org/abs/2410.18982

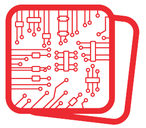
Recent posts



Understanding Reasoning LLMs

Methods and Strategies for Building and Refining Reasoning Models

FEB 5 · SEBASTIAN RASCHKA, PHD



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